# Anticipating State Action: Risk Perceptions and Consumption under Immigration Enforcement*

Alberto Ciancio†[1] and Camilo García-Jimeno‡[2]

[1]University of Glasgow

[2]Federal Reserve Bank of Chicago

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**Abstract**

We show that beliefs about government action shape economic behavior, and that government action shapes those beliefs—a two-way relationship we study in the context of U.S. immigration enforcement, which affects millions who make daily economic decisions under uncertainty about state action. Combining daily card-transaction records with arrest-level ICE data from 2014 to 2018, we document that enforcement follows predictable weekday patterns and that affected communities have learned them: consumption falls not only when raids occur but on the same weekday a week before and after, when none does—so behavior responds to anticipated risk, not realized enforcement, which studies keyed to events miss. Instrumenting for beliefs with this learnable structure, a ten-percentage-point rise in perceived risk lowers Hispanic foreign-born consumption by about 5 percent. A structural model of Bayesian learning and pent-up demand shows roughly half the decline is permanently foregone and half merely shifted to safer days. Because beliefs adjust slowly, the welfare effect of changing enforcement depends on how fast people learn the new regime: when enforcement eases, only 42 percent of the eventual gain materializes within a year.

**Keywords:** Immigration enforcement, Bayesian learning, beliefs, pent-up demand

**JEL Codes:** D83, D84, H11, J15, K37

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†alberto.ciancio@glasgow.ac.uk, 2 Discovery Pl, Glasgow G11 6EY, UK.

‡camilo.garcia-jimeno@chi.frb.org, 230 S. LaSalle Street, Chicago, IL.

# 1 Introduction

Governments use a variety of tools—censuses, standardized records, rule enforcement—to monitor and control populations. Populations, in turn, can respond to state action in multiple ways, creating a dynamic between state power and societal adaptation. In the Southeast Asian context, for example, James Scott has documented how populations employ everyday strategies—foot-dragging, evasion, dissimulation—to navigate state power and minimize exposure (Scott, 1985, 2009).[1]

This paper studies an informational variant in the context of U.S. immigration enforcement: we document how immigrant communities learn to anticipate Immigration and Customs Enforcement (ICE) operations, respond to *expected* enforcement (not just realized events), and adjust their daily behavior accordingly. Moreover, we quantify the consumption costs of this behavioral adaptation. By observing ICE's predictable weekday patterns and learning from experience, communities develop forecasting capacity. This adaptation partially mitigates—but does not eliminate—the costs of living under immigration enforcement risk.

Immigration enforcement affects far more than the individuals who are directly apprehended. By 2023, an estimated 26 million people in the United States lived in households with at least one unlawfully present immigrant (Pew Research Center, 2025). Media accounts describe them afraid to leave their homes, to go to work, or to seek medical care, and a growing academic literature documents these "chilling effects"—reductions in safety net participation, healthcare utilization, crime reporting, and labor force participation in response to enforcement activity (Alsan and Yang, 2024; Amuedo-Dorantes and Antman, 2022; East et al., 2023; Gonçalves et al., 2024; Watson, 2014). This literature typically estimates the effects of *realized* enforcement. The underlying behavioral mechanism, however, is rarely the shock itself—it is how the shock affects *beliefs* about enforcement risk.

This observation motivates four questions. First, do unexpected changes in immigration enforcement affect economic behavior? This is the standard question in the chilling-effects literature. Second, is immigration enforcement learnable? Learning is possible only if enforcement follows systematic patterns—across weekdays, locations, or over time. Third, do communities learn from experience and respond to expected enforcement risk? If individuals track enforcement patterns, behavior should adjust not only when raids occur, but also on days when raids are more likely to occur, even if no raid ultimately takes place. Finally, what are the welfare costs of immigration enforcement risk, and how does learning shape

[1] Acemoglu and Robinson (2019) go further, arguing that the competition dynamics between civil society and the state shape institutional development paths.

these costs when enforcement patterns change?

We address these questions in the context of routine federal immigration enforcement between 2014 and 2018, using two novel data sources: daily, transaction-level financial records from debit, credit, and prepaid cards, and arrest-by-arrest records of ICE interior enforcement obtained through FOIA litigation. The financial data allow us to measure consumption activity at the ZIP Code Tabulation Area (ZCTA) by day level, while the arrest records provide precise timing and location of enforcement events.

We begin with Operation Crosscheck—a coordinated, five-day enforcement surge in March 2015 that represented a sharp deviation from routine ICE activity. Because Crosscheck was not part of ICE's usual enforcement patterns, it constitutes an unexpected enforcement shock. We find that high–Hispanic foreign-born ZCTAs experienced large and immediate declines in consumption during Crosscheck, concentrated among populations most exposed to enforcement risk. This establishes that immigration enforcement has economically meaningful effects on consumption.

Having established that enforcement matters, we next ask whether enforcement is learnable. We show that ICE enforcement exhibits predictable patterns over time, with strong weekday effects and serial correlation. Once we condition on the day of the week and the previous day's enforcement activity, there is little remaining predictable structure. This implies that communities observing enforcement over time can form accurate forecasts, creating scope for learning.

We then turn to our central question: whether communities respond to expected enforcement risk rather than to realized enforcement alone. We design event studies centered on "clean" raid events—days when a raid occurred with no other raids in the surrounding $\pm 8$-day window. If communities learn enforcement patterns, consumption should decline not only on the raid day itself, but also on other high-risk days within the event window. Consumption is significantly depressed on the same weekday one week before and after the raid—even though, by construction, no raid occurred on those days—while it rises on low-risk weekdays within the window. These patterns provide direct evidence that behavior responds to beliefs about enforcement risk, not merely to realized raids.

Quantifying the causal effect of enforcement risk perceptions on economic activity requires addressing measurement error: realized raids imperfectly proxy for latent beliefs. With this purpose, we develop an instrumental variables strategy that exploits the learnable structure of enforcement: lagged same-weekday raids update beliefs without directly affecting current activity. The IV estimates reveal substantial attenuation in OLS and imply that a 10 percentage point increase in the perceived raid probability reduces Hispanic foreign-born economic activity by approximately 5 percent.

The weekday patterns we document—depressed activity on high-risk days, elevated activity on low-risk days—raise an identification question: is this consumption permanently foregone, or shifted inter-temporally to safer days? This distinction is crucial for welfare analysis, as reduced-form estimates cannot separately identify these margins. At the same time, enforcement patterns are not perfectly stable: relying on a structural break detection exercise we find 'enforcement regime' changes in roughly 30 percent of ZCTAs, primarily intensity shifts. This motivates a model where economic agents can adapt their learning when forecasts become unreliable.

We therefore estimate a structural model that incorporates Bayesian learning and pent-up demand incentives. In the model, suppressed activity can accumulate and partially rebound on safer days, and beliefs update with experience and can be discounted when past forecasts prove inaccurate. The model allows us to quantify the welfare costs of enforcement risk, distinguishing shifted from foregone consumption, and accounting for how learning frictions shape welfare when enforcement patterns change.

Three findings emerge. First, pent-up demand is significant: suppressed consumption generates above-baseline activity within 1–2 days (half-life of approximately 1.3 days). Second, we confirm the importance of chilling effects: higher perceived risk reduces activity. Third, individuals appear to discount unreliable beliefs, though this estimate is less precise. Impulse responses reveal roughly half the immediate chilling effect is recovered through bounce-back while half is genuinely foregone.

The estimated model supports counterfactuals that incorporate the full learning and pent-up demand dynamics: beliefs evolve endogenously as consumers observe enforcement signals, and suppressed demand accumulates and rebounds. The willingness to pay to eliminate enforcement risk entirely is 3.6 percent of Hispanic foreign-born consumption—but when enforcement actually stops, agents must learn this from their historical beliefs, and only 42 percent of the potential gain materializes within a year; the remaining 58 percent is the cost of learning frictions.

The policy-relevant direction, given the recent intensification of interior enforcement, is an increase in raid risk. We find that permanently doubling average enforcement risk reduces Hispanic foreign-born activity by 4 percent once beliefs have adjusted to the new regime—on the order of $3 billion per year in foregone earnings nationally. Because beliefs adapt only gradually, this steady-state figure is an upper bound on the short-run loss from a comparable surge: the realized response builds up over time as communities learn that enforcement has changed. These magnitudes are directly relevant to the roughly threefold rise in arrests over 2025–2026 relative to our sample period.

Under perfect foresight about enforcement, activity falls by 1.7 percent on average—immigrants tend to underestimate raid risk—with the largest responses where forecasts had been least accurate. A budget-neutral move to uniform enforcement intensity across ZCTAs has small effects (mean gains under 0.6 percent), indicating limited welfare distortion in ICE's current spatial allocation. Finally, adaptive discounting of unreliable beliefs raises economic activity by about 1.8 percent on average, with larger gains where enforcement is harder to predict.

Our paper makes several contributions: First, we interpret community learning about governmental action as a form of informational adaptation to navigate state power. We cleanly separate responses to *expected* enforcement (event studies with the day $\pm 7$ effect) from responses to *unexpected* enforcement (Operation Crosscheck), showing that individuals react to novel information, and can anticipate enforcement through learning. We quantify short-run 'chilling' effects on economic activity of enforcement risk perceptions. We additionally provide suggestive evidence of learning sophistication: economic agents adjust the importance they give to their learned beliefs when forecasts prove unreliable. Finally, our framework allows us to distinguish consumption that is merely *shifted* (timing distortion) from consumption that is permanently *foregone*, a distinction important for welfare analysis and elusive in previous empirical work. It also allows us to quantify the value of information about enforcement risk, and to measure the dynamic implications of several immigration enforcement policies. More broadly, our framework speaks to a recurring challenge in the economics of enforcement: most causal estimates recover behavioral responses *within* a fixed enforcement regime (specific deterrence), whereas what matters for policy is often the effect of *changing* the regime (general deterrence) (Eeckhout et al., 2010). By embedding learning and dynamic adjustment, our structural model bridges the two.

Our study contributes to several areas of research. First, as mentioned above, a growing strand of literature documents that immigration enforcement activity generates "chilling effects" on both unauthorized and lawfully present immigrants, including U.S. citizens (Watson and Thompson, 2022). We contribute to this literature by documenting chilling effects along another margin—consumption—and, importantly, by isolating immigrants' *beliefs* about enforcement risk and estimating the causal impact of beliefs on behavior.

Our paper also relates to the literature studying the economic costs of living under the threat of state action. Studies of conflict (Tapsoba (2023); Besley and Mueller (2012)), terrorism (Alfano and Görlach (2022); Abadie and Gardeazabal (2003); Draca et al. (2011)), and policy uncertainty (Pastor and Veronesi (2012)) demonstrate that expectations of future state actions depress asset values, investment, and human capital accumulation, even in the absence of direct exposure. In particular, in the economics of crime literature, deterrence is determined by the perceived probability of punishment (Chalfin and McCrary, 2017). While this literature highlights the importance of perceived risk, beliefs are typically proxied by realized events or latent risk estimated by the researcher. An exception is Lochner (2007), who uses self-reported arrest probabilities from the NLSY97 to show that deterrence depends on the perceived likelihood of arrest and that individuals update these beliefs based on their own experiences. Closely related is García-Jimeno (2016), who studies learning about law enforcement under alcohol prohibition and shows that beliefs about enforcement effectiveness play a central role in shaping compliance and political outcomes. We contribute by explicitly modeling belief formation and learning about enforcement risk and by isolating the effect of these beliefs, rather than realized enforcement, on consumption.

We draw on the adaptive learning framework of Evans and Honkapohja (2001) and Orphanides and Williams (2005), where agents learn about inflation dynamics or policy rules in macroeconomic models. Adaptive learning models were first estimated in macroeconomic settings by Milani (2007) in a DSGE framework, while Slobodyan and Wouters (2012) introduced constant-gain learning, which allows the speed of learning to vary over time—an important feature when beliefs respond strongly to salient events. More recently, Carvalho et al. (2023) show that agents can recognize the un-anchoring of inflation expectations through sophisticated forms of adaptive learning. Similarly, using microdata, Malmendier and Nagel (2016) show that individuals learn about inflation through experience, with more recent experiences receiving greater weight. We contribute to the adaptive learning literature by studying belief formation in a setting where agents learn about a latent enforcement regime from realized enforcement events rather than from aggregate macroeconomic outcomes. Unlike most of this literature, we estimate beliefs using micro-level consumption behavior and high-frequency enforcement data.

# 2 Data and Institutional Context

## 2.1 Immigration Enforcement in the United States

About 5 percent of the population residing in the U.S. is unlawfully present, having entered without inspection or having overstayed their visas. Most participate in the labor force (5.6 percent of the workforce in 2023). Three quarters originate from Latin America and concentrate in California, Texas, and Florida (Pew Research Center, 2025).

Immigration enforcement operates through two channels: border enforcement by Customs and Border Protection (CBP), and interior enforcement by Immigration and Customs Enforcement (ICE). We focus on interior enforcement—arrests made within communities, not at ports of entry. ICE operates through 24 regional Areas of Responsibility (AORs), each covering multiple states. Because the federal government has broad discretion over enforcement, intensity varies across AORs and over time—variation central to our empirical approach.

Between 2014 and 2018, the period we study, three quarters of interior enforcement involved ICE taking custody of individuals already detained by local law enforcement.[2] ICE, however, can also arrest individuals directly in the community—the focus of this paper. Though fewer in number, these arrests are highly visible and disruptive. So-called immigration raids—large-scale operations arresting hundreds simultaneously, such as Operation Crosscheck in March 2015—have received extensive media coverage and generated widespread fear.[3] Raids take place in workplaces, parking lots, and outside homes and immigration courts.[4] In practice, most ICE arrests come not from large operations but from daily targeted operations with one or a few targets[5]. Throughout the paper, we use the term "raid" to describe community-based ICE arrest operations, including those involving a single target.

We study the period from October 2014 through May 2018, straddling two administrations with different enforcement priorities. The Obama administration emphasized enforcement against those with serious criminal convictions; the Trump administration expanded enforcement efforts to a broader population. This policy shift, combined with daily data, generates variation in both the level and predictability of enforcement. Figure 1 displays weekly interior arrests over the sample period, with vertical lines marking key policy events.

## 2.2 Data

**Immigration Enforcement.** Data on ICE arrests come from the Transactional Records Access Clearinghouse (TRAC) at Syracuse University, which obtained arrest-by-arrest records through FOIA litigation. We use arrests classified as "community (at large)"—those outside jails or prisons. Each record includes the arrest date, county[6], and most serious criminal

[2]See TRAC (https://tracreports.org/phptools/immigration/arrest/). These transfers occur largely through the Secure Communities program, under which inmates' fingerprints are shared with ICE; local cooperation is not mandatory and has been politically contested (Ciancio and García-Jimeno, 2024).

[3]For example, "Immigrant Communities in Hiding: 'People Think ICE Is Everywhere,'" *The New York Times*, January 30, 2025, and "ICE raids are looming. Panicked immigrants are skipping work, hiding out and bracing for the worst," CNN, July 12, 2019.

[4]Under the Immigration and Nationality Act, ICE officers may arrest noncitizens with or without a warrant under specified conditions (INA §§ 236, 287; 8 U.S.C. §§ 1226, 1357), but absent consent or exigent circumstances generally need a judicial warrant to enter a private residence.

[5]These targeted operations sometimes lead to collateral arrests, arrests of individuals who were not the original targets of an operation but are encountered during its execution and determined to be candidates for removal by the officers in charge.

[6]ICE does not officially track arrest locations; TRAC reconstructed county information through hundreds of FOIA requests and cross-referencing multiple sources. In some cases, the county designation includes surrounding areas where exact location could not be isolated.

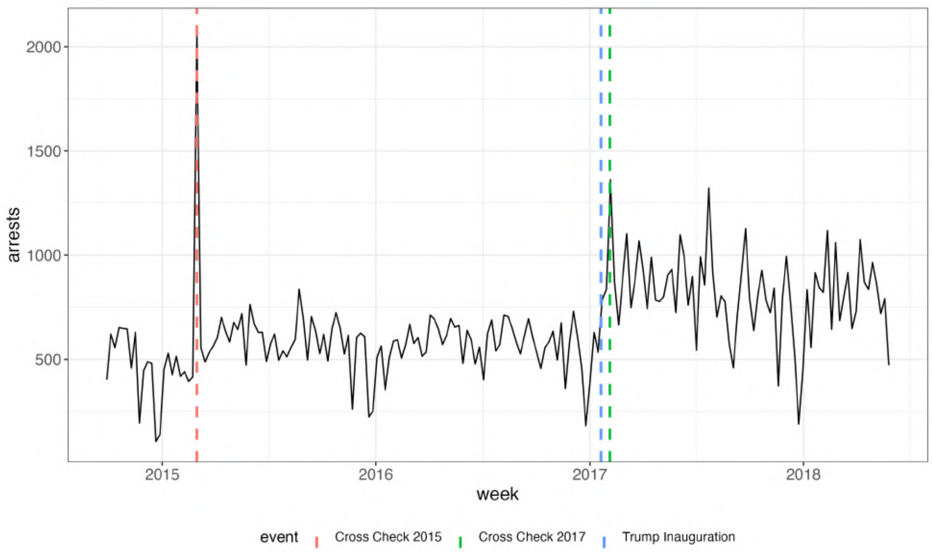



**Figure 1:** Weekly ICE Interior Arrests, October 2014 – May 2018.

*Notes:* Weekly count of ICE interior raid arrests from TRAC data. Red and green vertical lines indicate Operations Crosscheck (March 2015 and January 2017) and President Trump's inauguration (January 2017). Interior arrests exclude arrests by CBP and transfers from other agencies.

conviction (categorized as level 1, 2, or 3, or none). The most common categories are no conviction (28%), driving under influence (16%), traffic offense (5%), assault (4%), drug possession (4%) and illegal re-entry (4%).

Because our consumption data are at the ZIP Code Tabulation Area (ZCTA) level, we map county arrests to ZCTAs using Census relationship files. ZCTAs can span multiple counties; we associate a ZCTA with each county containing at least 2.5 percent of its population, then sum arrests across associated counties.[7] The resulting data include 17,765 ZCTAs with at least one arrest during the sample period. Figure A1 displays the geographic distribution of arrests.

**Economic Activity.** We measure economic activity using data from Facteus, a financial data aggregator that provides transaction-level records from a very large sample of debit, credit, and prepaid cards. We keep accounts with at least one transaction both before and after our sample period—holding the set of accounts fixed over time—and aggregate to the ZCTA level.

Our main outcome is the *share of accounts active*—accounts with at least one transaction in the ZCTA-day, divided by total accounts. This captures the extensive margin: whether individuals transact at all, rather than how much they spend. The extensive margin suits

[7] We provide details of this mapping procedure in Appendix A.

| | Below Median HF | | Above Median HF | | |
|---|---|---|---|---|---|
| | Mean | Std. Dev. | Mean | Std. Dev. | $p$-value |
| *Panel A: Economic Activity* | | | | | |
| Share accounts active | 0.105 | 0.031 | 0.106 | 0.028 | 0.206 |
| Spent amount ($) | 1,756 | 4,306 | 2,526 | 5,375 | 0.000 |
| Number of accounts | 129.0 | 112.7 | 183.4 | 140.4 | 0.000 |
| Accounts per 1,000 pop. | 8.263 | 3.500 | 7.038 | 3.383 | 0.000 |
| *Panel B: Demographics* | | | | | |
| Hispanic foreign-born share | 0.006 | 0.005 | 0.078 | 0.079 | 0.000 |
| Median household income ($) | 65,787 | 28,124 | 65,897 | 28,164 | 0.824 |
| Population | 16,394 | 12,138 | 27,665 | 17,920 | 0.000 |
| *Panel C: ICE Enforcement* | | | | | |
| Any raid (daily) | 0.067 | 0.148 | 0.185 | 0.242 | 0.000 |
| Raid count (cond. on raid) | 1.401 | 0.655 | 1.873 | 1.097 | 0.000 |
| Ever raided | 0.635 | — | 0.841 | — | 0.000 |
| N (ZCTAs) | 6,433 | | 6,432 | | |
| N (Days) | 1,339 | | | | |

**Table 1:** Summary Statistics

*Notes:* Summary statistics by median Hispanic foreign-born (HF) share. Panel A reports economic activity from Facteus. Panel B reports ZCTA characteristics from 2018 ACS 5-Year estimates. Panel C reports ICE enforcement from TRAC. Last column reports $p$-values for difference of means tests. Sample period: October 2014 – May 2018 (1,339 days).

avoidance behavior: if people stay home, they make fewer transactions.[8] In robustness exercises we also use the total monetary value of the transactions. The balanced panel contains approximately 30,000 ZCTAs observed daily. We provide additional details on these data and the construction of our sample in Appendix A.1.

**Demographics.** ZCTA-level characteristics come from the 5-Year 2018 American Community Survey. Our key variable is the *Hispanic foreign-born share*—the fraction both Hispanic and foreign-born—which proxies for the population most exposed to enforcement risk. We also use median household income, total population, and racial composition.

**Sample Construction and Summary Statistics.** The final sample merges the three data sources at the ZCTA-date level. We impose a "selection cone" requiring at least 30

[8]Our data do not distinguish online from in-person transactions. Because online activity entails no physical exposure, it is unlikely to respond to enforcement risk. Its inclusion may only attenuate the measured avoidance, so our estimates understate the true in-person response.

accounts and 3–25 accounts per 1,000 population, ensuring adequate coverage while excluding implausible penetration rates. This yields 12,865 ZCTAs over 1,339 days (17.2 million observations). Of these, 73.8% experienced at least one arrest ("ever-raided"). Table 1 presents summary statistics split by median Hispanic foreign-born share (1.7%). The share of accounts active averages 10.5%, similar across groups. High-Hispanic-foreign-born ZCTAs experience more enforcement: 18.4% of days have at least one raid versus 6.7%, and 84.1% are ever-raided versus 63.5%.

# 3 Beliefs and Patterns of Immigration Enforcement

Throughout, we focus on *perceived enforcement risk*: community members' expectation that immigration enforcement will occur in their location on a given day. This risk is not directly observed by the researcher; it may be shaped by learning from past enforcement activity or by information transmitted through social, organizational, or institutional channels. Our interest is in how this risk shapes behavior: a *chilling effect* is the causal response of economic activity to changes in it—not to enforcement itself. Absent variation in risk perceptions, chilling effects cannot be measured. Consider a linear model of economic activity:

$$y_{z,t} = \alpha + \beta r^e_{z,t} + \epsilon_{z,t} \tag{1}$$

where $r^e_{z,t} \equiv \mathbb{E}[r_{z,t}|\mathcal{I}_{z,t}]$ is the subjective probability of a raid in community $z$ on day $t$, $\mathcal{I}_{z,t}$ is information available at the beginning of the period, and $\beta$ captures the chilling effect. We assume $\mathbb{E}[\epsilon_{z,t} r_{z,t} \mid \mathbf{x}_{z,t}] = 0$ for an appropriate set of controls $\mathbf{x}_{z,t}$—ICE's enforcement decisions do not respond to daily economic shocks. Because we aggregate data at the ZCTA level, we also assume beliefs are common within communities. If we observed beliefs, regressing activity on them with ZCTA and time fixed effects would recover $\beta$.

## 3.1 Operation Crosscheck

We begin studying Operation Crosscheck, a coordinated enforcement surge that provides clean identification of responses to *unexpected* enforcement. It consisted of a coordinated five-day enforcement action by ICE from March 1–5, 2015, resulting in 2,059 arrests nationwide.[9] Unlike routine enforcement, this was a planned surge executed simultaneously across all ICE

[9] See https://www.dhs.gov/news/2015/03/09/2059-criminals-arrested-ice-nationwide-operation. Operation Crosscheck is part of ERO's National Fugitive Operations Program, targeting "priority" populations (e.g., at-large criminals with violent convictions or gang members). Earlier Crosscheck operations predate our sample; a second, in January 2017, overlaps with President Trump's first inauguration and we do not use it.

districts, and establishes a baseline chilling effect: surprise enforcement reduces short-run economic activity in locations with large numbers of immigrants.

**Crosscheck as an Information Shock.** Crosscheck is valuable as a natural experiment because it represents *unexpected* enforcement. Three features distinguish it from routine enforcement. First, the operation was planned nationally and executed simultaneously—timing was not predictable from local conditions. Second, Crosscheck began on a Sunday and ran through Thursday, deviating from the typical weekday concentration we will document in Section 3.2. Third, the scale was unprecedented: while typical enforcement involves 1.2 arrests, Crosscheck-affected ZCTAs saw dozens simultaneously. Because communities could not have anticipated Crosscheck from past patterns, any response represents reaction to unexpected news rather than pre-planned avoidance.

**Event Study Design.** Let event time dummies be

$$D_{z,t}^{k} = \mathbf{1}[\tau_{z,t} = k], \qquad k \in \{-K, \ldots, +K\}, \tag{2}$$

and consider the baseline event study specification

$$y_{z,t} = \alpha_z + \delta_t + \sum_{k \neq 0} \gamma_k D_{z,t}^{k} + \varepsilon_{z,t} \tag{3}$$

where $y_{z,t}$ is economic activity, $\alpha_z$ are ZCTA fixed effects, and $\delta_t$ are date fixed effects. The coefficients $\gamma_k$ capture mean differences in the outcome $k$ days from baseline $k = 0$. In practice, our focus is on whether economic activity changes differentially in communities with larger fractions of people subject to immigration enforcement risk, so our specification allows for heterogeneity along the Hispanic foreign born margin, $h_z$: $\gamma_k = \gamma_k^0 + \gamma_k^1 h_z$.

Crosscheck represents a surprise signal $S_\tau$ that expands the information set to $\mathcal{I}_{zt} \cup S_\tau$ for periods $\tau > 0$, inducing a belief innovation. Denoting with $^*$ variables after partialling out fixed effects, the least squares estimator identifies

$$\gamma_k = \beta \underbrace{\left(\mathbb{E}\left[r_{zt}^{e*}(\mathcal{I}_{z,t} \cup S_\tau) \mid \tau = k\right] - \mathbb{E}\left[r_{zt}^{e*}(\mathcal{I}_{z,t}) \mid \tau = 0\right]\right)}_{\text{Belief innovation from surprise}} + \underbrace{\left(\mathbb{E}\left[\epsilon_{zt}^{*} \mid \tau = k\right] - \mathbb{E}\left[\epsilon_{zt}^{*} \mid \tau = 0\right]\right)}_{\text{Residual difference}} \tag{4}$$

where expectations are over ZCTAs, with $\tau$ indexing event time. Under the assumption that Crosscheck timing is exogenous to local consumption shocks, the residual difference is zero, and the coefficients $\gamma_\tau$ recover the chilling effect $\beta$ scaled by the average belief innovation.

Before the signal arrives, $S_\tau = \varnothing$, so we expect flat pre-trends followed by a dip when the information set expands.

**Empirical Specification.** We construct an event study sample with clean windows. We include ZCTAs that (i) experienced at least one arrest during Crosscheck (March 1–5, 2015), and (ii) had no arrests in the 10 days before or after. The second requirement ensures that other enforcement actions do not contaminate the event study windows. We measure event time relative to Tuesday, March 3—the middle of the Crosscheck week. This yields 802 ZCTAs with clean Crosscheck events.

We use the two-stage approach of Gardner et al. (2024) for heterogeneous treatment effects in staggered designs. In the first stage, we estimate ZCTA-by-weekday and date fixed effects using untreated observations (ZCTA-days without raids in a ten-day window):

$$y_{zt} = \alpha_{z \times d(t)} + \delta_t + \varepsilon_{zt} \tag{5}$$

where $d(t) \in \{\text{Mon}, \ldots, \text{Sun}\}$ denotes the weekday. The ZCTA-by-weekday effects absorb systematic day-of-week patterns, ensuring we identify the Crosscheck effect from the enforcement shock rather than regular weekly cycles. Because we estimate them on untreated observations only, treatment does not pollute these fixed effects. And because we have a very large sample of untreated ZCTAs and days, we can estimate them precisely.

In the second stage, we regress the residualized outcome $\tilde{y}_{zt} = y_{zt} - \hat{\alpha}_{z \times d(t)} - \hat{\delta}_t$ on the interactions between the Hispanic foreign-born share $h_z$ and event time indicators, or more flexibly, with dummies for the deciles of the Hispanic foreign born share distribution. We base our inference on 1,000 bootstrap replications clustered at the ZCTA level.

**Identification.** The ZCTA-by-weekday fixed effects absorb regular consumption patterns and the predictable component of enforcement (Section 3.2); what remains is Crosscheck's surprise, so we need not model beliefs explicitly here.

**Results.** Figure 2 presents coefficients on the interaction terms on the event-time by Hispanic foreign-born share. The coefficients show a sharp negative dip on Monday and Tuesday (event time $-1$ and 0)—the core of the operation—indicating that ZCTAs with higher Hispanic foreign-born shares experienced reduced economic activity during the enforcement surge. The effect is sizable: for the average ZCTA in the Crosscheck sample, consumption drops by about 8% of baseline activity during the peak of the operation. Communities with larger at-risk populations curtailed activity in response to unexpected enforcement.

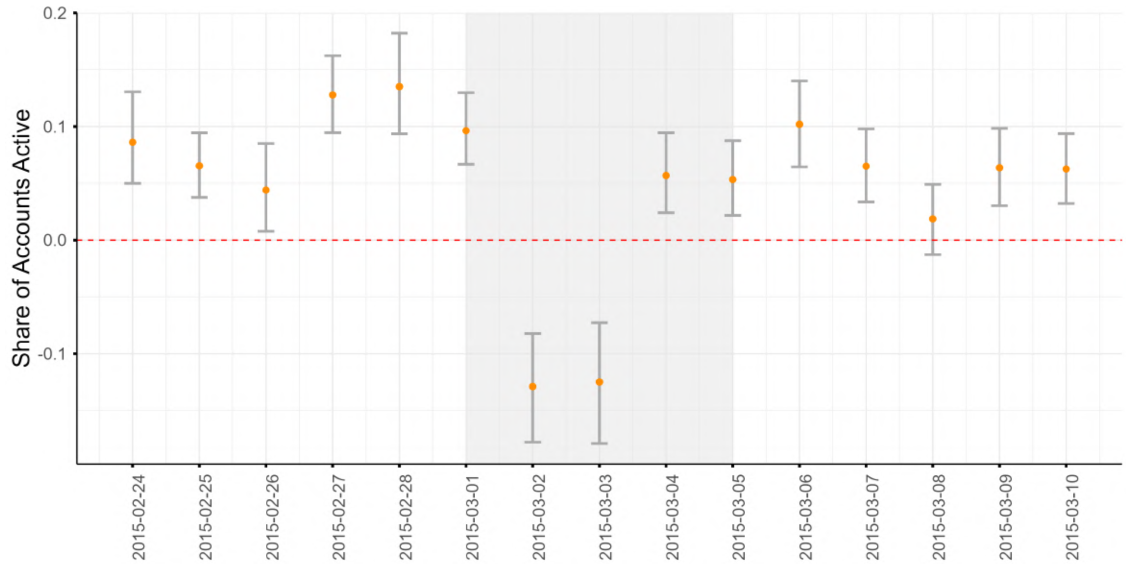



**Figure 2:** Operation Crosscheck Event Study: Hispanic Foreign-Born Share Interactions

*Notes:* Event study coefficients on the interaction between event time indicators and Hispanic foreign-born share. The sample includes 802 ZCTAs that experienced a raid during Operation Crosscheck (March 1–5, 2015) with no raids in the 10 days before or after. Event time is relative to March 3, 2015 (Tuesday of Crosscheck week). The estimation uses the Gardner et al. (2024) two-stage method with ZCTA × weekday and date fixed effects. Error bars show 95 percent confidence intervals from 1,000 Bayesian bootstrap replications (Rubin, 1981) clustered at the ZCTA level.

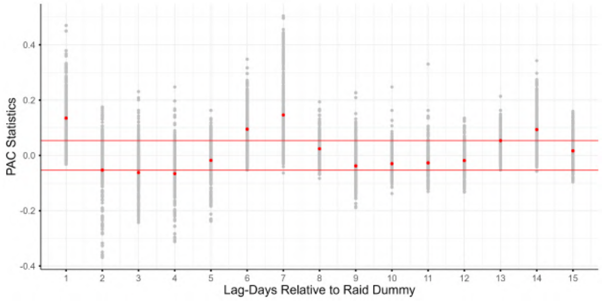
Figure A4 presents results by Hispanic foreign-born share decile. The negative effect on Monday and Tuesday concentrates in top deciles: deciles 1–3 show clear, statistically significant dips, while deciles 5–9 remain flat around zero. This suggests fear among at-risk populations, rather than some channel affecting all communities equally.

The Crosscheck analysis establishes an important baseline: unexpected enforcement reduces immediately subsequent economic activity in immigrant communities. Whether communities also respond to *expected* enforcement depends on whether ICE is predictable—the question we take up next.

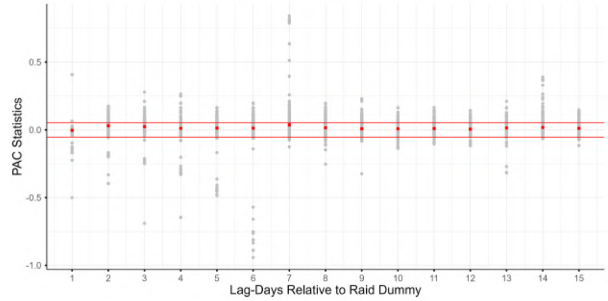
## 3.2 ICE Enforcement Patterns

**Weekday Structure.** ICE enforcement follows pronounced day-of-the-week patterns. Figure 3a displays the partial autocorrelation function (PACF) of the raid indicator, showing strong autocorrelation at lags that are multiples of seven days. ICE operations concentrate on weekdays rather than weekends, and these patterns persist within ZCTAs over time.

These weekday patterns vary across ZCTAs. Figure A5 displays a heatmap of weekday enforcement profiles, with each row representing one ZCTA and columns showing the share of raid-days on each weekday. We group ZCTAs by peak enforcement day and sort




**(a)** All observations




**(b)** Residualized

**Figure 3:** Partial Autocorrelation of ICE Raids

*Notes:* Panel (a) shows the partial autocorrelation function (PACF) of the raid indicator. Panel (b) shows the PACF of the residualized raid indicator, after controlling for weekday dummies and a lagged raid indicator. Red points indicate mean PACF values across ZCTAs; gray points show individual ZCTA values.

them by concentration within each group. While mid-week enforcement is generally higher (visible as the vertical structure), intensity varies across ZCTAs (visible as the horizontal variation). Some ZCTAs have concentrated enforcement on a single day; others have diffuse profiles. This suggests communities cannot simply adopt the aggregate pattern, and must learn their own local enforcement rhythms. The weekday structure likely reflects operational constraints: ICE officers work regular schedules, courts and processing facilities operate on weekday hours, and multi-day operations may be planned around the work week.

**Serial Correlation.** Beyond weekday patterns, enforcement exhibits strong serial correlation. A raid today increases the probability of a raid tomorrow, even controlling for weekday. This likely reflects multi-day operations—when ICE targets an area, officers conduct arrests over several days before moving on. This creates an additional learnable signal.

**Residual Autocorrelation** Can individuals grasp predictable patterns using a simple forecasting rule? Figure 3b displays the partial autocorrelation function (PACF) of the raid indicator—after regressing it on weekday dummies and the lagged raid indicator. The autocorrelation is flat: once we control for weekday and the first lag, no predictable structure remains. Thus, two features describe ICE enforcement: (1) weekday patterns varying across ZCTAs, and (2) serial correlation captured by the first lag. A learner conditioning on weekday and yesterday's outcome can form accurate forecasts, as long as patterns remain stable.

## 3.3 Event Studies: Inferring Beliefs about Enforcement Risk

Do communities actually learn ICE's weekday patterns? We answer this with event studies centered on "clean" raid events: days when a raid occurred with no other raids in the surrounding $\pm 8$ day window. By centering on realized raids, we anchor on days when enforcement risk was high (since raids cluster on preferred weekdays). If communities track ICE patterns, consumption should dip not only on the raid day itself, but on all high-likelihood days within the window—including days $-7$ and $+7$, where by construction no raid occurred.

**Event Study Design.** The event-time dummies $D^k_{z,t} = \mathbf{1}[\tau_{z,t} = k], k \in \{-8, \dots, +8\}$ are now defined relative to a day $\tau = 0$ when $r_{z,\tau} = 1$, and the baseline specification is the same as in (3). The coefficients $\gamma_k$ now capture the average difference in economic activity $k$ days away from a 'high enforcement risk' day, relative to the baseline day.

The least squares estimator of the $\boldsymbol{\gamma} = \{\gamma_k\}$'s from (3) identifies

$$\gamma_k = \beta \underbrace{\left(\mathbb{E}[r^{e*}_{z,t} \mid \tau = k] - \mathbb{E}[r^{e*}_{z,t} \mid \tau = 0]\right)}_{\text{Belief gradient in event time}} + \underbrace{\left(\mathbb{E}[\epsilon^{*}_{z,t} \mid \tau = k] - \mathbb{E}[\epsilon^{*}_{z,t} \mid \tau = 0]\right)}_{\text{Residual}} \tag{6}$$

where the expectations are taken over the cross-section of ZCTA-date pairs, with $\tau$ indexing event time. The key insight is that by centering event windows on realized raids ($r_{z,t} = 1$ at $\tau = 0$), we align clocks across ZCTAs at times when beliefs about enforcement risk are plausibly elevated. Because ICE follows weekday patterns, day 0 is likely a high-probability weekday for that ZCTA, making days $\pm 7$ (the same weekday) also high-probability, and different weekdays, say $\pm 4$, lower-probability. This converts idiosyncratic weekday patterns into a common event-time gradient that we can estimate.

Without anchoring on realizations, beliefs would have no systematic reason to vary with event time, so anchoring generates the identifying variation. While Crosscheck recovers responses to belief innovations from surprises, the clean raid event studies here recover the cross-weekday belief gradient.

**Temporal Alignment between Raid Risk and Economic Activity.** As with Crosscheck, we use "clean" windows where a raid occurred at $\tau = 0$ with no raids in $[-8, +8]$. Our sample contains 33,091 clean raid events across 5,967 ZCTAs (562,547 ZCTA-day observations; Appendix Figure A6 shows the geographic distribution). We implement the Gardner et al. (2024) two-stage procedure, allowing $\gamma_k$ to vary with the Hispanic foreign-born share.

Figure 4 presents our central finding. Orange points show consumption coefficients along the Hispanic foreign-born share gradient. The blue line overlays the average raid likelihood by event day from an autocorrelation regression (see Appendix Figure A7). The correspondence is striking: consumption closely mirrors ZCTA-level enforcement risk. When the raid likelihood is high (days $-7$, 0, $+7$), consumption is depressed; when the likelihood is low

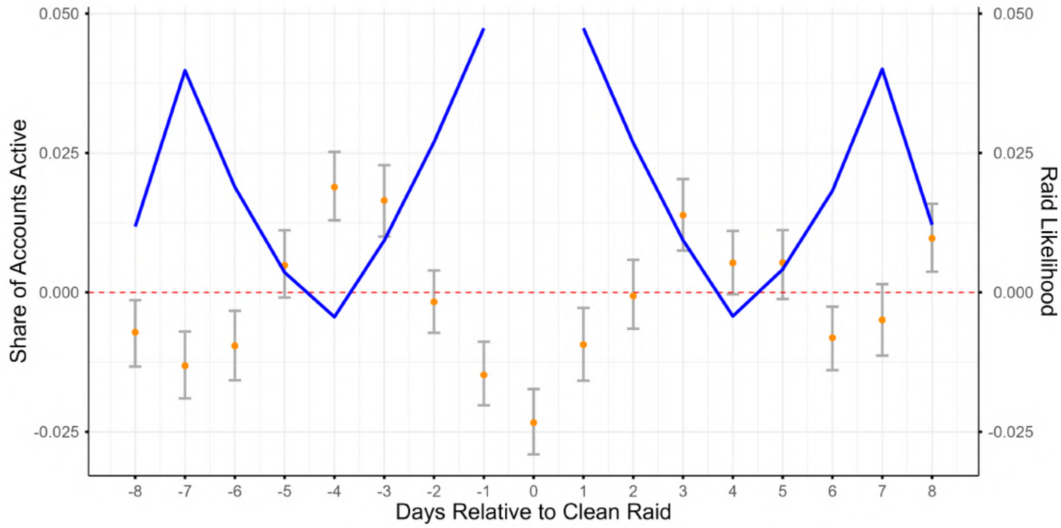



**Figure 4:** Clean Raids Event Study: Hispanic Foreign-born Share Interactions

*Notes:* Consumption coefficients (orange, left axis) overlaid with raid likelihood (blue, right axis) by days relative to clean raid. The event-study specification includes ZCTA and date fixed effects. The event study coefficients are those on the interaction between event time and the Hispanic foreign-born share. Error bars show 95 percent confidence intervals from 1,000 Bayesian bootstrap replications (Rubin, 1981) clustered at the ZCTA level.

(days −4, +4), consumption recovers.

These patterns indicate that economic activity responds to *perceived enforcement risk*, not to realized enforcement alone: consumption is depressed even on the same weekday one week before and after a raid, when by construction no enforcement occurs. These anticipatory responses are consistent with belief formation but do not pin down a single mechanism—learning from past enforcement, transmission through social or organizational networks, institutional coordination (e.g., employer scheduling), or some combination.

**Robustness.** We also estimate the belief gradient by Hispanic foreign-born share decile (Figure A8). The seven-day pattern is strongest in the highest deciles, attenuates in deciles 4–6, and disappears in the bottom deciles. Effects concentrate exactly where unlawfully present populations are likely to be the largest, supporting our interpretation. For this reason, we restrict our subsequent analysis (IV and structural estimation) to ZCTAs in the top quintile of the Hispanic foreign-born share.

Another concern about our results is that the weekly pattern could reflect weekday-specific consumption (e.g., people shop more on Saturdays) rather than risk perceptions. Date fixed effects absorb aggregate weekday effects but not ZCTA-specific patterns. Formally, if $\epsilon_{zt} = \theta_z^{d(t)} + u_{zt}$ where $\theta_z^{d(t)}$ captures ZCTA-specific weekday effects, the residual in (6) becomes $\mathbb{E}[\theta_z^k] - \mathbb{E}[\theta_z^{d(0)}]$. Since $k = \pm 7$ share the same weekday as $k = 0$, we have

$\mathbb{E}[\theta^{\pm 7}] = \mathbb{E}[\theta^0]$. But for other weekdays, $\mathbb{E}[\theta_z^k]$ could vary, generating a seven-day pattern even with $\beta = 0$.

We address this by replacing ZCTA fixed effects with ZCTA $\times$ weekday fixed effects, absorbing $\theta_z^{d(t)}$. These fixed effects difference out each ZCTA's *average* perceived risk for a given weekday—the component of beliefs that recurs every seven days—so the gradient is now identified purely from transitory, within-weekday deviations around that average. A realized raid updates beliefs and marks a spell of elevated enforcement, pushing perceived risk on nearby days above what is usual for that weekday; it is this variation, not the permanent differences in risk across weekdays, that the event-time profile traces. Appendix Figure A9 shows the overlay plot with these fixed effects. The pattern survives but attenuates, which is not surprising since the fixed effects absorb the large recurring component of the belief gradient and retain only its transitory part. The pattern is also robust to using log spending as the outcome (Appendix Figure A10).

## 3.4 Instrumental Variable Strategy

The event study shows that consumption tracks expected raid risk but the event study coefficients do not directly identify the chilling effects coefficient $\beta$ from equation 1. In this section, we show how to recover an average chilling effect when there is learning about immigration enforcement patterns.

**Realizations as Proxies for Beliefs.** Studies of enforcement effects typically estimate responses to enforcement shocks. But behavior responds to *beliefs* about risk, not shocks themselves, creating a measurement error problem: the realized shock is a noisy proxy for the latent belief.[10] To formalize this, recall from equation (1) that economic activity responds to beliefs if $\beta \neq 0$. Because beliefs are unobserved, a natural approach is to use the realized raid indicator $r_{zt}$ as a proxy for $r^e_{zt}$. Define the prediction error as $\tilde{\eta}_{zt} \equiv r_{zt} - r^e_{zt}$—the gap between the raid realization and what agents expected. Substituting $r^e_{zt} = r_{zt} - \tilde{\eta}_{zt}$ into the model yields:

$$y_{zt} = \alpha_z + \delta_t + \beta r_{zt} + (\epsilon_{zt} - \beta \tilde{\eta}_{zt}) \tag{7}$$

The composite error now includes the prediction error $\tilde{\eta}_{zt}$, which is mechanically correlated with the regressor $r_{zt}$. Under the maintained assumption that $\text{cov}(r_{zt}, \epsilon_{zt}) = 0$ conditional

[10] In other contexts—such as the effect of interest rate changes on consumption—both the shock itself and beliefs about future shocks affect outcomes, creating an inherent identification challenge. Here, the consumption decision must be made before observing whether a raid occurs, so only beliefs matter at decision time, a convenient feature of our setting.

on fixed effects, the OLS estimand from (7) takes the form (see Appendix E for derivation):

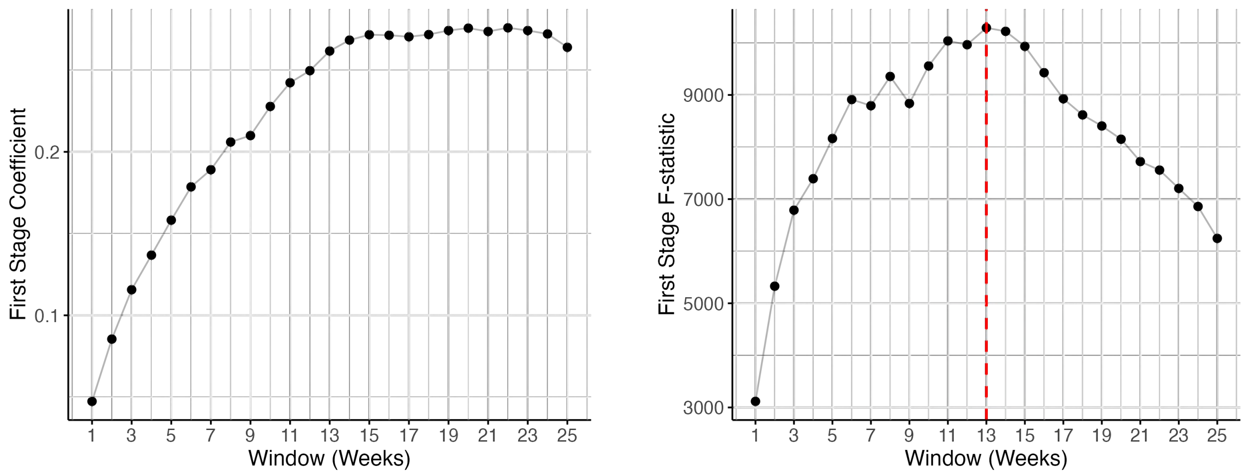
$$\beta_{OLS} = \beta \cdot [\mathbb{E}[r^e_{zt}|r_{zt} = 1] - \mathbb{E}[r^e_{zt}|r_{zt} = 0]] \tag{8}$$

The bracketed bias factor measures how much beliefs differ between raid and non-raid days, and it scales the OLS estimand by belief quality. At the extremes, OLS identifies nothing when beliefs are uninformative ($r^e_{zt}$ uncorrelated with $r_{zt}$) and is unbiased when agents forecast raids perfectly ($r^e_{zt} = r_{zt}$). The empirically relevant case lies in between: with informative but imperfect beliefs, OLS is attenuated toward zero but recovers the correct sign whenever beliefs carry some information ($\text{cov}(r^e_{zt}, \mathbb{P}(r_{zt} = 1 \mid r^e_{zt})) \geq 0$; see Appendix E).

**Using Learning to Construct Instruments.** Learning suggests a solution: if individuals form beliefs from past enforcement, lagged enforcement—on the same weekday—should shift beliefs without directly affecting current consumption. We construct an instrument as follows. For each ZCTA-weekday-date, we compute the rolling mean of raid indicators over the previous $\ell$ weeks *on the same weekday*. The identifying assumption underlying this strategy is that lagged same-weekday enforcement affects current economic activity only through its effect on *perceived enforcement risk*. Because people decide whether to go out before observing that day's enforcement, behavior responds to expectations rather than to realized raids, and recent enforcement shifts those expectations even when no raid occurs. Conditional on location and weekday fixed effects, lagged same-weekday enforcement therefore moves activity only through perceived risk, not through any direct effect on contemporaneous activity.

Our IV strategy relies on the idea that agents form beliefs $r^e_{zt}$ using past information, so lagged realizations of the signal $r_{zt-k}$ belong to $\mathcal{I}_{z,t-1}$ and predict $r^e_{zt}$. It does not require informationally optimal beliefs (i.e., $r^e_{zt} = \mathbb{E}[r_{zt}|\mathcal{I}_{z,t}]$). Instead, three high-level conditions are sufficient: (i) relevance: $\text{Cov}(r^e_{zt}, r_{z,t-k}) \neq 0$ for some lag $k$ (e.g., same-weekday $k = 7$); (ii) lag-orthogonality of the belief–realization gap: $\mathbb{E}[\eta_{zt}|\mathcal{I}_{z,t-1}] = 0$; and (iii) outcome exogeneity: $\mathbb{E}[\epsilon_{zt}|\mathcal{I}_{z,t-1}] = 0$. Under (i)–(iii), the population moment $\mathbb{E}[r_{z,t-k}(y_{zt} - \beta r_{zt})] = 0$ holds, so 2SLS using $r_{z,t-k}$ identifies $\beta$. This weaker orthogonality is similar to that from applied-micro "learning from experience" designs that use lagged own/peer realizations as belief shifters (e.g., Conley and Udry (2010); Munshi (2004)), while avoiding assumptions of fully optimal Bayesian updating. In our pooled panel with date and ZCTA by weekday fixed effects, we use same-weekday lags (e.g., $t - 7$, $t - 14$) as instruments.

**Instrument Selection.** Figure 5 displays first-stage statistics across window lengths, $\ell$. Given our findings above, we restrict the sample here to ZCTAs in the top quintile of the




**(a)** First Stage Coefficient

**(b)** First Stage F-statistic

**Figure 5:** Instrument Selection: First-Stage Statistics by Window Length

*Notes:* First-stage statistics from IV regressions using the rolling mean of lagged same-weekday raids as instrument for the current raid indicator. Panel (a) shows first-stage coefficient; panel (b) shows first-stage F-statistic. The vertical dashed line in panel (b) indicates the optimal window (13 weeks) by F-statistic. All regressions include ZCTA $\times$ weekday and date fixed effects. Standard errors clustered by ZCTA $\times$ weekday. The sample is restricted to ZCTAs in the top quintile of the Hispanic foreign-born share distribution.

Hispanic foreign-born share distribution. Panel (a) shows the first-stage coefficient; Panel (b) shows the F-statistic. The first-stage relationship exhibits a clear pattern as a function of lag length window: the coefficient increases up to around 13 weeks and flattens thereafter, while the F-statistic shows an inverted-U shape that peaks at 13 weeks. This inverted-U is consistent with adaptive learning that weights recent signals more heavily; static or permanent information mechanisms, or responses whose salience fades, would not generate it. All specifications yield F-statistics well above conventional thresholds (the smallest exceeds 3,000), so weak instruments are not a concern. We use the 13-week window as our baseline.

**Results.** Table 2 presents IV and OLS estimates. Column (1) shows OLS with ZCTA $\times$ weekday and date fixed effects: the coefficient is negative but very small and statistically insignificant ($-0.0001$, s.e. $= 0.0002$). As expected: ZCTA $\times$ weekday fixed effects absorb the predictable component of beliefs, leaving only noise and driving OLS toward zero.

Column (2) presents the IV estimate using the 13-week rolling mean with the same fixed effects. The coefficient increases to $-0.007$, statistically significant at the 5% level. The correction is large but sensible since the realized raid indicator is a crude proxy—it equals one only when a raid occurs, even though beliefs may be elevated on any historically high-probability day.

The remaining columns probe the sensitivity of this estimate to the fixed effects structure

| | (1) OLS | (2) IV | (3) IV | (4) IV | (5) IV Overid |
|---|---|---|---|---|---|
| *Panel A: Second Stage* | | | | | |
| Raid Indicator | -0.000 | -0.007 | -0.012 | -0.036 | -0.009 |
| | (0.000) | (0.003) | (0.002) | (0.005) | (0.003) |
| *Panel B: First Stage* | | | | | |
| Instrument (13-week avg) | | 0.286 | 0.425 | 0.592 | |
| | | (0.009) | (0.007) | (0.006) | |
| Raid Lag 1 Week | | | | | 0.051 |
| | | | | | (0.002) |
| Raid Lag 2 Weeks | | | | | 0.044 |
| | | | | | (0.002) |
| ZCTA × Weekday FE | Yes | Yes | | | Yes |
| ZCTA FE | | | Yes | | |
| Weekday FE | | | Yes | | |
| Date FE | Yes | Yes | Yes | Yes | Yes |
| Observations | 1,490,112 | 1,490,112 | 1,490,112 | 1,490,112 | 1,582,050 |
| $R^2$ | 0.56 | 0.56 | 0.52 | 0.18 | 0.56 |
| First-stage F-stat | | 13,377 | 36,078 | 95,517 | 3,784 |

**Table 2:** Effect of Raid Beliefs on Economic Activity

*Notes:* The dependent variable is the share of accounts active in a ZCTA-day. The raid indicator equals one if at least one ICE arrest occurred in the ZCTA on that day. Standard errors clustered by ZCTA × weekday in parentheses. Columns 2–4 instrument for the raid indicator using the 13-week rolling mean of lagged same-weekday raids. Column 5 uses 1-week and 2-week same-weekday lags as separate instruments. Sample restricted to ZCTAs in the top quintile of the Hispanic foreign-born share distribution.

and instrument specification. Column (4) omits ZCTA and weekday controls entirely, including only date fixed effects; the IV coefficient jumps to $-0.036$, much larger than our baseline. This inflation highlights the importance of accounting for cross-sectional and cross-weekday heterogeneity in enforcement exposure that the instrument otherwise picks up.

Indeed, once we include ZCTA and weekday fixed effects in any form, the estimate stabilizes. Column (3) uses additive (non-interacted) ZCTA, weekday, and date fixed effects; the coefficient is $-0.012$, similar to the baseline. Column (5) returns to the interacted ZCTA × weekday specification but uses an over-identified model with 1-week and 2-week same-weekday lags as separate instruments; the coefficient is $-0.009$, again in the same range. The consistency across columns (2), (3), and (5)—whether the ZCTA and weekday effects are interacted or additive, whether the model is exactly identified or over-identified—suggests that the key requirement is controlling for location-specific weekday patterns in some form, not the precise specification of those controls.

**Interpretation.** The IV estimates recover chilling effects operating through expectations about enforcement risk. Whether perceived risk evolves through individual learning from past enforcement, information transmission within communities, or institutional coordination does not affect the interpretation of the IV estimates as long as past enforcement activity modifies communities' information sets.

These estimates imply that the mean effect of raid risk perceptions on economic activity is large: if they increase by 10 percentage points (e.g., an increase from 0.10 to 0.20, roughly a doubling relative to the mean raid rate) in a ZCTA at the median Hispanic foreign-born share among our high-exposure sample (approximately 14 percent), the share of active accounts falls by $0.007 \times 0.1 = 0.07$ percentage points. This implies a consumption reduction of approximately 5 percent among the Hispanic foreign-born.[11] Back of the envelope calculations translate this into \$10M per day in foregone earnings.[12]

Studies using realized enforcement shocks may substantially understate responses to enforcement *risk*. The bias is not endogeneity but measurement error: a mismatch between what we observe (shocks) and what drives behavior (beliefs). Our IV strategy relies on this learning: lagged enforcement, and its weekday structure in particular, is informative about current risk perceptions.

# 4 Model

The reduced-form analysis showed that consumption responds to expected raids indicating that communities have learned enforcement patterns and act on their beliefs. Economic activity tracks beliefs about immigration enforcement, dipping on high-risk days and bouncing back on low-risk days. This pattern could reflect, however, permanently foregone consumption or inter-temporal substitution across days, with very different welfare implications. To distinguish these margins from each other and from how they are mediated by changes in risk perceptions, we propose a simple structural model with two key ingredients: i) a pent-up demand motive, and ii) learning about immigration enforcement risk based on past enforce-

[11] If the consumption fall is only among the Hispanic foreign born, then for this group it falls by $0.07/0.14 = 0.5$ percentage points. This is 5 percent of the 10 percent of daily accounts active average in the sample.

[12] We obtain this result in 3 steps. 1: Divide by $h_z$. Since $\hat{\beta} = h_z \cdot \hat{\beta}^H$, the H-type-specific coefficient is about $-0.05$ at the median $h_z = 0.14$ in our high-exposure sample. 2: A 10 pp belief shock reduces H-type going-out by 0.5 pp. Applied to 13.5M H-type employed workers nationally, that's about 67,500 people staying home on a given day. 3: At roughly \$150/day in average earnings for this population, this is about \$10.1M per day in foregone earnings. The main caveat here is that we are counting only the extensive margin (whether they use electronic payments at all).

| $\pi^N(a,r)$ | **Action** | |
|---|---|---|
| **Raid** | $a_{izt}=0$ | $a_{izt}=1$ |
| $r_{zt}=0$ | $\pi_{SN}$ | $\pi_{ON}$ |
| $r_{zt}=1$ | $\pi_{SN}$ | $\pi_{ON}$ |

N-type payoffs

| $\pi^H(a,r)$ | **Action** | |
|---|---|---|
| **Raid** | $a_{izt}=0$ | $a_{izt}=1$ |
| $r_{zt}=0$ | $\pi_{SN}$ | $\pi_{ON}$ |
| $r_{zt}=1$ | $\pi_{SR}$ | $\pi_{OR}$ |

H-type payoffs

Table of ex-post payoffs from the tracking game.

ment patterns, both drivers of consumption dynamics. We specify how agents make daily going-out decisions trading off the benefits of current activity under risk against the dynamic cost of accumulating unsatisfied pent-up demand. We then describe how agents form and update beliefs about enforcement risk based on observed raids. The model allows us to separately identify the parameters governing i) the response of risk perceptions on economic activity —chilling effects—, ii) belief formation and adaptive responses, and iii) pent-up demand dynamics.

## 4.1 Going-Out Decisions with Pent-Up Demand

**Individual Problem.** We consider a community $z$ consisting of a continuum of individuals $i$ of two types $\tau$: those at risk of immigration enforcement, $H$, and those not subject to it, $N$. The fraction of $H$ types is $h_z$. At the beginning of every day $t = 1, 2, ...$, each person decides whether to 'go out' ($a_{izt} = 1$) or 'stay home' ($a_{izt} = 0$). Typically, going out implies engaging in economic activity—work, study, business, etc. The individual payoffs from this choice depend on type, and on whether there is an immigration enforcement action $r_{zt}$, i.e., a raid, later in the day in the community. We summarize per-period payoffs with a utility function that mimics a 'tracking problem' (e.g., Baley and Veldkamp (2023)). For $i \in \tau$,

$$U^{\tau}_{izt}(a, r) = \pi^{\tau}(a, r) + \tilde{\xi}^a_t + \tilde{\zeta}^a_{zd(t)} + \epsilon^a_{izt}, \tag{9}$$

where $\tilde{\xi}^a_t$ and $\tilde{\zeta}^a_{zd(t)}$ are day and community-by-weekday common components, while the $\epsilon^a_{izt}$ are idiosyncratic unobservables, i.i.d. over $i$. The table below presents the type-specific component of payoffs:

Because $N$-types are not subject to immigration enforcement, their payoff does not depend on whether a raid takes place or not, only on whether they stay home or go out. We will maintain $\pi_{ON} > \pi_{SN}$, so $N$-types only choose to stay home as a function of their idiosyncratic shock $(\epsilon^0_{izt}, \epsilon^1_{izt})$. The problem is more interesting for $H$-types. They would like to coordinate their choice with the government's enforcement action. In the absence of a raid, they would also rather go out (up to the idiosyncratic shock). If there is a raid, however, it is natural to assume they prefer to stay in, so $\pi_{SR} > \pi_{OR}$.

At the time of taking the action people do not know whether a raid will take place that day. However, they have a belief $B_{zd(t)} = \mathbb{P}(r_{zt} = 1|\mathcal{I}_{zt})$, common to everyone in the community, that a raid will happen. $\mathcal{I}_{zt}$ denotes the information available to the community at the beginning of the day. Note that we allow this belief to be weekday specific given our earlier discussion. This belief will only matter among $H$ types since only they face immigration enforcement risk.

**Pent-up demand.** Staying home today may raise the value of going out later, so a present reduction in activity can be partly or fully compensated by higher activity in subsequent days (see Beraja and Zorzi (2024) for a similar mechanism with durables). A chilling decline driven by a high raid belief may thus be undone by above-average activity later. To allow for this we introduce a dynamic component of preferences. Each person accumulates 'pent-up demand for going out', $x_{izt}$. When $x > 0$, there is a backlog of unfulfilled desire to go out. We capture this idea modeling the evolution of pent-up demand as:

$$x_{izt+1} = (1-\delta)x_{izt} - (a_{izt} - \kappa). \tag{10}$$

$\delta \in (0,1)$ governs the rate at which pent-up demand decays over time, and $\kappa > 0$ is the desired long-run frequency of going out. Throughout we set $\kappa = 1$, so that staying in at time $t$ generates an increase in pent-up demand of one day.[13]

Accumulated pent-up demand generates an instantaneous convex flow utility cost:

$$c(x_{izt}) = \frac{\psi}{2}x_{izt}^2, \qquad \psi \geq 0. \tag{11}$$

When $\psi = 0$ there is no cost of having refrained from going out in the past, and the going-out decision is static: low present consumption does not lead to higher future consumption. For $\psi > 0$, staying home today increases pent-up demand, making this a dynamic discrete choice problem. In this case, the comparison of payoffs when going out or staying home must take into account that staying home today, say because the raid belief is high, increases $x_{izt+1}$, which is costly next period.

**Dynamic Programming Problem.** We can express a type-$\tau$ person's dynamic problem at the beginning of period $t$ with the following Bellman equation:

[13] $\kappa = 1$ has two implications: i) pent-up demand converges to $1/\delta$ for a person that never goes out. ii) for $x_{iz0} \geq 0$, pent-up demand cannot be negative, that is, people cannot find themselves 'ahead of schedule'.

$$
\begin{aligned}
V^\tau(x_{izt}) =& \mathbb{E}_\epsilon\Bigg[\max_{a_{izt}\in\{0,1\}}\Bigg\{\mathbb{E}_r\Bigg[a_{izt}U^\tau_{izt}(1,r)+(1-a_{izt})U^\tau_{izt}(0,r)\Bigg|\mathcal{I}_{zt}\Bigg] \\
&- \frac{\psi}{2}x^2_{izt}+\beta V^\tau\big((1-\delta)x_{izt}+\kappa-a_{izt}\big)\Bigg\}\Bigg].
\end{aligned} \tag{12}
$$

This is a standard dynamic discrete choice problem. Following Hotz and Miller (1993), assuming the idiosyncratic payoff components $\epsilon^a_{izt}$ are i.i.d. type-I extreme value-distributed implies choice probabilities of the form

$$
\mathbb{P}(a_{izt}=1|x_{izt},B_{zd(t)},\tau)=\frac{\exp(v^\tau_1(x_{izt},B_{zd(t)})-v^\tau_0(x_{izt},B_{zd(t)}))}{1+\exp(v^\tau_1(x_{izt},B_{zd(t)})-v^\tau_0(x_{izt},B_{zd(t)}))} \tag{13}
$$

where we can write the value differences as

$$
v^\tau_1(x_{izt},B_{zd(t)})-v^\tau_0(x_{izt},B_{zd(t)})=\alpha_0+\alpha_1 B_{zd(t)}\mathbf{1}[\tau=H]+\xi_t+\zeta_{zd(t)}+\Delta^\tau(x_{izt})
$$

for $\tau\in\{H,N\}$, and $\xi_t\equiv\tilde{\xi}^1_t-\tilde{\xi}^0_t$, $\zeta_{zd(t)}\equiv\tilde{\zeta}^1_{zd(t)}-\tilde{\zeta}^0_{zd(t)}$, $\alpha_0\equiv\pi_{ON}-\pi_{SN}>0$, $\alpha_1\equiv(\pi_{OR}-\pi_{SR})-(\pi_{ON}-\pi_{SN})$, and

$$
\Delta^\tau(x_{izt})\equiv\beta\big[V^\tau((1-\delta)x_{izt}-(1-\kappa))-V^\tau((1-\delta)x_{izt}+\kappa)\big]. \tag{14}
$$

Given the acute costs of being arrested for H types, absent any shocks it is natural to expect the net gain from going out relative to staying home if there is a raid, $\pi_{OR}-\pi_{SR}$, to be negative. And since the net gain from going out relative to staying home if there is no raid, $\pi_{ON}-\pi_{SN}$, is positive, $\alpha_1<0$. The component $\Delta^\tau(x_{izt})$ captures the dynamic utility gain from reduced pent-up demand into next period when going out instead of staying home.[14]

**Aggregation and Mixture Structure.** In practice we work with data aggregated at the ZCTA level, so we estimate a version of the model above with a representative agent for each type, each of whom keeps track of the aggregate pent-up demand of his group, $x^\tau_{zt}$. Thus, observed economic activity is

$$
A_{zt}=h_z\Lambda^H_{zt}\left(w^H_{zt}\right)+(1-h_z)\Lambda^N_{zt}\left(w^N_{zt}\right) \tag{15}
$$

[14]The stationarity assumption shuts down anticipatory substitution—front-loading before expected high-risk days—which would require a non-stationary value function depending on the entire path of beliefs. As a result, our estimates attribute all compensating dynamics to pent-up demand, so $\psi$ may absorb some timing variation and overstate the pent-up cost. Relaxing stationarity would separate backward-looking catch-up from forward-looking substitution, at significant computational cost and limited added identification given the regularity of weekly patterns. Model validation, thus, is important to assess the empirical relevance of the pent-up demand mechanism.

where $\Lambda(w)$ is the logistic function, and

$$w_{zt}^{\tau} = \alpha_0 + \alpha_1 B_{zd(t)} \mathbf{1}[\tau = H] + \Delta^{\tau}(x_{zt}^{\tau}) + s_{zt}, \quad s_{zt} \equiv \xi_t + \zeta_{zd(t)}.$$

The two latent stocks evolve as in (10) separately for H and N types. Notice that the date and weekday payoff shocks are common to both groups. The groups' aggregate choices differ through the impact of beliefs and the differential evolution of their aggregate pent-up demand. Equation (15) is a mixture logistic regression with observed weights, in beliefs, the dynamic pent-up component, and time and weekday fixed effects. We can estimate this regression solving for the value functions for each group as long as we can build a measure of beliefs.

## 4.2 Learning

### 4.2.1 Bayesian Learning about Enforcement Risk

We model how communities form and update beliefs about immigration enforcement risk. Our specification rests on two maintained assumptions. First, ICE's enforcement strategy does not depend on the community's decisions—neither economic activity nor belief updating affects when or where ICE conducts operations. Second, communities observe successful raids (those resulting in arrests) but not failed enforcement attempts. This assumption ensures that agents' information sets coincide with ours, which is necessary for constructing beliefs from observable data.

Given that raid outcomes are binary, we adopt a Beta-Bernoulli learning framework. Our reduced-form evidence motivates two refinements to standard Bayesian updating. First, because ICE enforcement exhibits pronounced day-of-the-week patterns, we allow agents to maintain weekday-specific beliefs: a Monday raid updates beliefs about future Mondays, not about Tuesdays. Second, because raids exhibit positive serial correlation—a raid yesterday strongly predicts a raid today—we allow beliefs to condition on whether a raid occurred the previous calendar day. Together, these refinements mean agents effectively maintain fourteen parallel learning processes: one for each weekday $\times$ previous-day-raid-status combination.

Formally, let $a_d^k$ and $b_d^k$ denote the Beta parameters for weekday $d \in \{1, \ldots, 7\}$ conditional on previous-day raid status $r_{t-1} = k \in \{0, 1\}$. The prior mean for weekday $d$ following $r_{t-1} = k$ is

$$B_{d(0)}^k = \frac{a_{d(0)}^k}{a_{d(0)}^k + b_{d(0)}^k}.$$

Standard Beta-Bernoulli updating implies that after observing the same-weekday outcome

$r_{t-7}$ from one week earlier, the parameters evolve as

$$\begin{bmatrix} a^{r_{t-1}}_{d(t)} \\ b^{r_{t-1}}_{d(t)} \end{bmatrix} = \begin{bmatrix} a^{r_{t-1}}_{d(t-7)} \\ b^{r_{t-1}}_{d(t-7)} \end{bmatrix} + \begin{bmatrix} \mathbf{1}\{r_{t-7} = 1\} \\ 1 - \mathbf{1}\{r_{t-7} = 1\} \end{bmatrix}. \tag{16}$$

The posterior mean belief about a raid on day $t$, conditional on weekday $d(t)$ and previous-day outcome $r_{t-1}$, is

$$B_{zd(t)} \equiv \mathbb{P}(r_t = 1 \mid r_{t-1}, r_{t-7}) = \frac{a^{r_{t-1}}_{d(t)} + \mathbf{1}[r_{t-7} = 1]}{a^{r_{t-1}}_{d(t)} + b^{r_{t-1}}_{d(t)} + 1}.$$

With learning incorporated, $B_{zd(t)}$ is no longer a free parameter but a deterministic function of the observed raid history $\{r_s\}_{s<t}$ and the initial prior.

### 4.2.2 Regime Changes and Belief Sensitivity

The updating rule in (16) implicitly assumes individuals perceive ICE's strategy to be stationary. However, enforcement patterns may shift over time as ICE reallocates resources or adjusts priorities. To study this, we apply the Bai and Perron (2003) structural break detection method to weekday-specific raid probabilities, testing for significant changes in these probabilities within ZCTAs (more details in appendix section F). Figure A11 displays detected break dates. We identify 513 breaks in 439 ZCTAs, and no breaks for the remaining ZCTAs. The distribution spikes in January 2017 (around President Trump's first inauguration), but regime changes occurred throughout the sample. We also find that the weekday coefficients in the new regimes are highly correlated with those in the old regime, i.e. the ranking of days by risk is preserved (Figures A13a and A13b). Regime changes primarily reflect intensity shifts, not re-shuffling.

In such an environment, agents may prefer to weight recent signals more heavily than distant ones (see, e.g., Ma et al. (2023) in the computational cognitive science literature context, or the burgeoning literature in macroeconomics that studies expectation formation: Carvalho et al. (2023); Cho and Kasa (2015); Weber et al. (2025)).[15] We address this concern not by modifying the learning rule directly—which would introduce difficult-to-identify gain

[15] One could accommodate this with an updating rule of the form

$$\begin{bmatrix} a^{r_{t-1}}_{d(t)} \\ b^{r_{t-1}}_{d(t)} \end{bmatrix} = \lambda_t(\overline{f}_t) \begin{bmatrix} a^{r_{t-1}}_{d(t-7)} \\ b^{r_{t-1}}_{d(t-7)} \end{bmatrix} + \begin{bmatrix} \mathbf{1}\{r_{t-7} = 1\} \\ 1 - \mathbf{1}\{r_{t-7} = 1\} \end{bmatrix},$$

where the gain parameter $\lambda_t < 1$ underweights past information as a function of previous forecast errors $f_t$. This is akin to the distinction between constant-gain and decreasing-gain learning in the macroeconomic expectations literature.

parameters—but through a belief-uncertainty interaction component described below, which attenuates behavioral responses when past beliefs have proven unreliable.[16]

As beliefs evolve through learning, agents accumulate a track record of forecast accuracy. We define the cumulative average forecast error as a weighted average of absolute forecast errors up to time $t$:

$$\overline{f}_{zt} = \frac{1}{t-1}\sum_{s=1}^{t-1} \omega_{st}|r_{zs} - B_{zd(s)}|, \tag{17}$$

where $\omega_{st}$ are exponential weights that down-weight older signals. $\overline{f}_{zt}$ captures how well agents' beliefs have tracked realized enforcement over the sample period. Notice that $\overline{f}_{zt}$ is *not* weekday-specific: we expect individuals to base their perceptions about regime changes on an overall assessment of forecast accuracy across all days, not on weekday-specific tallies.

A natural hypothesis is that the strength of the chilling effect varies with agents' confidence in their beliefs. When past forecasts have been accurate, agents should weight their beliefs heavily in decision-making. When forecasts have systematically missed the mark—perhaps because enforcement patterns shifted unexpectedly—agents may respond more cautiously to their current beliefs. To capture this, we allow the belief coefficient to be heterogeneous in forecast accuracy:

$$\alpha_1(\overline{f}_{zt}) = \alpha_{11} + \alpha_{12} \cdot \overline{f}_{zt}. \tag{18}$$

The H-type latent utility index thus becomes $w^H_{zt} = \alpha_0 + \left(\alpha_{11} + \alpha_{12} \cdot \overline{f}_{zt}\right) \cdot B_{zt} + c(x^H_{zt}) + s_{zt}$.

The parameter $\alpha_{11}$ captures the average effect of beliefs on the propensity to go out; we expect $\alpha_{11} < 0$ since higher perceived raid probability should reduce economic activity. The parameter $\alpha_{12}$ governs how this effect varies with forecast accuracy. If $\alpha_{12} > 0$, agents respond more weakly to their beliefs when past forecasts have been poor: the total belief effect $(\alpha_{11} + \alpha_{12} \cdot \overline{f}_{zt})$ becomes less negative as $\overline{f}_{zt}$ rises. This specification nests the baseline model: when $\alpha_{12} = 0$, the belief effect is constant at $\alpha_{11}$.

This mechanism generates a sharp prediction: following structural breaks in enforcement patterns, we should observe spikes in $\overline{f}_{zt}$ as the break induces a mismatch between agents' learned beliefs and the new enforcement regime. We test this prediction directly in Section 5.4.

[16]The gain parameter is notoriously weakly identified: it affects observables only indirectly through beliefs (so the likelihood is often flat in the gain dimension), it is collinear with the persistence of structural shocks, and beliefs are themselves generated regressors—issues well known in the macroeconomic adaptive-learning literature (Milani, 2007; Slobodyan and Wouters, 2012).

### 4.2.3 Discussion: Mis-specification of the Learning Model

A natural concern with any structural learning model is whether the belief-formation process is correctly specified.[17] Our specification addresses four ways actual learning might deviate from a simple Bernoulli rule: *non-stationarity* in ICE's strategy, which we accommodate by letting the belief response vary with the cumulative forecast error (dampening responses when accumulated errors signal a regime shift); *serial dependence*, via separate tallies conditional on the previous day's raid; *weekday heterogeneity*, via weekday-specific tallies; and *signal noise*—we observe only successful raids, but so do agents, which keeps our information set consistent with theirs and is conservative, since any additional information would only sharpen beliefs.

# 5 Estimation and Results

## 5.1 Sample and Estimation Strategy

Our estimation sample consists of the 352 ZCTAs in the top quintile of the Hispanic foreign-born share distribution that experienced at least 30 raids during the period (Oct. 2014 to May 2018). This threshold ensures sufficient within-ZCTA variation in beliefs and forecast errors. The resulting panel contains 471,328 ZCTA-day observations spanning 1,339 days.

We initialize beliefs using predicted raid probabilities from a probit regression estimated on a burn-in period of the 105 initial days of our sample. These burn-in observations are excluded from the estimation sample, ensuring beliefs and forecast errors have stabilized. The probit includes interactions of weekday and previous-day-raid status with Hispanic share and log median household income, as well as ICE Area of Responsibility fixed effects. Appendix G.2 provides more details and reports the probit estimates (Table A2).

We estimate the structural parameters $\boldsymbol{\theta} = (\psi, \delta, \alpha_{11}, \alpha_{12})$ by matching model-implied consumption to observed consumption, concentrating out the high-dimensional fixed effects. The approach exploits the invertibility of the mixture model: given $\boldsymbol{\theta}$, initial conditions for the prior $B_{z0}$, and for the pent-up demand state $x_{z0}$, and observables $(A_{zt}, B_{zt}, x_{zt}, h_z)$, we can recover a unique implied shock $s_{zt}$ that rationalizes each observation. The estimation proceeds in three steps. First, for a candidate $\boldsymbol{\theta}$, we solve for the stationary value functions $V^\tau(x)$ and compute the dynamic incentive term $\Delta^\tau(x)$ from equation (14). Second, we invert the mixture model to recover the shocks $\{s_{zt}(\boldsymbol{\theta})\}$ that rationalize observed consumption.[18] Third, we regress these recovered shocks on date and ZCTA-by-weekday fixed effects. The

[17] See Manski (2004) for a discussion of these conceptual issues.

[18] This inversion yields a unique solution, as we describe in subsection G.3.

estimator $\hat{\boldsymbol{\theta}}$ minimizes the residual sum of squares of this regression.

We fix the discount factor $\beta = 0.995$ (daily data) and the pent-up demand target rate $\kappa = 1$, and use an exponential function for the forecast error weights $\omega_s$. The fixed effects absorb the intercept $\alpha_0$, so we cannot separately identify it. Appendix G provides complete details on the solution algorithm, inversion procedure, and numerical implementation.

## 5.2 Identification

**Dynamic parameters.** The pent-up cost $\psi$ and depreciation rate $\delta$ are identified from serial correlation in consumption after accounting for the fixed effects, following periods with changes in beliefs. When $\psi = 0$, the model collapses to myopic behavior: a belief shock $\Delta B_t$ affects only consumption on day $t$. When $\psi > 0$, pent-up states evolve smoothly, creating inter-temporal substitution (bounce-back). Thus, negative auto-correlation in economic activity —over calendar time— following a change in $B_{zd(t)}$ informs $\psi$. The positive auto-correlation in economic activity that follows while pent-up demand is still high then informs $\delta$, since this parameter governs the horizon of its decay.[19]

**Belief parameters.** The belief effect $\alpha_{11}$ is identified from cross-sectional variation in the Hispanic foreign-born share $h_z$ interacted with beliefs $B_{zt}$. High-$h$ communities experience larger consumption drops one and seven days after a raid. The interaction parameter $\alpha_{12}$ is identified from time-varying sensitivity to beliefs within communities: periods of high vs. low forecast accuracy generate different consumption responses to the same belief level.

Notice that the belief effect and the pent-up effect work in *opposite directions* after a raid shock that shifts beliefs: while the increased risk perception causes consumption to drop (if $\alpha_{11} < 0$), accumulated pent-up demand causes consumption to rise later. This timing difference—immediate suppression followed by delayed recovery—allows separate identification of the two mechanisms.

## 5.3 Main Results

### 5.3.1 Main estimates

Table 3 presents the parameter estimates with three sets of standard errors: from the numerical Hessian, and Newey-West HAC with 8 and 30 lags.

[19] In contrast with previous literature on pent-up demand, where typically the depreciation parameter is set externally (e.g., Beraja and Zorzi (2024)), the variation available in our data together with the parsimony of the model allows us to estimate $\delta$.

| Parameter | Estimate | SE (Std) | SE (HAC-8) | SE (HAC-30) |
|---|---|---|---|---|
| $\psi$ | 2.23 | 0.01 | 0.09 | 0.09 |
| $\delta$ | 0.42 | 0.00 | 0.01 | 0.01 |
| $\alpha_{11}$ | -0.28 | 0.02 | 0.12 | 0.12 |
| $\alpha_{12}$ | 0.93 | 0.04 | 0.52 | 0.51 |

**Table 3:** Pent-up Demand Model Parameter Estimates

*Notes:* 434,368 observations across 352 ZCTAs. SE (Std) reports standard errors from the numerical Hessian. SE (HAC-8) and SE (HAC-30) report Newey-West heteroskedasticity and autocorrelation-consistent standard errors with 8 and 30 lags, respectively.

**Belief effect ($\alpha_{11}$).** The negative estimate $\hat{\alpha}_{11} = -0.28$ confirms that higher perceived raid risk reduces economic activity. A 10 percentage point increase in beliefs $B_{zt}$ decreases the Hispanic foreign born 'going out' probability by 0.7 percentage points (evaluated at the sample mean).

**Belief-uncertainty interaction ($\alpha_{12}$).** The positive estimate $\hat{\alpha}_{12} = 0.93$ indicates that agents respond less strongly to their beliefs when past forecasts have been inaccurate. The total belief effect is $(\alpha_{11} + \alpha_{12} \cdot \bar{f}_{zt}) \cdot B_{zd(t)}$. At the mean cumulative forecast error ($\bar{f} \approx 0.07$), the effective belief coefficient is $-0.28 + 0.93 \times 0.07 = -0.21$, about 25 percent weaker than the baseline effect. At the 90th percentile of forecast error ($\bar{f} \approx 0.12$), the effective coefficient is $-0.28 + 0.93 \times 0.12 = -0.16$, nearly 40 percent weaker. The interaction mechanism provides a behavioral foundation for why chilling effects may be heterogeneous across time and communities.

**Pent-up demand ($\psi$, $\delta$).** The estimates $\hat{\psi} = 2.233$ and $\hat{\delta} = 0.417$ imply meaningful inter-temporal substitution. When chilling effects suppress consumption, pent-up demand accumulates and eventually manifests as above-baseline activity. The depreciation rate implies a half-life of $\log(2)/\log(1/(1-0.417)) \approx 1.3$ days, suggesting that pent-up effects dissipate relatively quickly. Figure A14 in Appendix G.1 displays the estimated value function and the dynamic incentive term $\Delta(x)$, illustrating how the option value of going out varies with the pent-up state.

**Fixed effects ($\xi, \zeta$).** The estimation concentrates out 1,234 date fixed effects and 2,464 ZCTA-by-weekday fixed effects. Figure A17 in Appendix G displays these estimated fixed effects: the date effects capture aggregate time-series variation (holidays, macroeconomic shocks) and reveal a downward sloping trend in aggregate activity over time in the sample, while the ZCTA-by-weekday effects illustrate meaningful level differences in aggregate activity between ZCTAs across weekdays.

**Implied beliefs and forecast errors.** Figure A18 displays the implied beliefs, cumulative forecast errors, and time-varying marginal effect of beliefs. Beliefs exhibit the expected weekday structure, with higher raid probabilities on weekdays than weekends. The distribution of marginal effects of beliefs, $\alpha_{11} + \alpha_{12} \cdot \bar{f}_{zt}$, shows mean shifts at different points in time but is overall stationary. The most pronounced mean shifts take place around the times when there is a large number of ZCTAs for which we detected regime changes. This time variation mirrors the decreasing-gain term in adaptive-learning models and is a quantitatively important source of variation in economic activity. Figure A19 in Appendix G.8 additionally illustrates implied mean raid probabilities and the evolution of beliefs for a handful of ZCTAs.

### 5.3.2 Impulse Response Analysis

We now trace the model's dynamic implications through a series of impulse-response exercises. Starting from steady state, we trace how economic activity responds to an unexpected increase in beliefs—the dynamic analog of the chilling effect. We do so by simulating 'going out' choices forward from steady state under two scenarios: i) a shock to beliefs followed by immediate reversion back to baseline, and ii) a shock to beliefs after which beliefs are allowed to evolve forward in response to the shock. The impulse response measures the percent difference in economic activity relative to steady state. Figure 6 presents the first scenario—immediate reversion—under a 15 percentage point shock to beliefs at $t = 0$; Appendix Figure A20 presents the second, in which beliefs evolve forward after the shock. This is a large shock, as estimated mean beliefs outside weekends are between 6 and 10 percent. We report results on mean overall economic activity, and on mean economic activity among the Hispanic foreign born.

On impact, the belief shock causes economic activity to fall, with a much larger effect for the Hispanic foreign-born group (approximately 1 percent of their economic activity) than for the aggregate (approximately 0.14 percent of overall economic activity). This differential reflects that only the H-type group responds to enforcement beliefs and that the Hispanic foreign born are 14 percent of the typical ZCTA population in our estimation sample.

Following the initial drop, pent-up demand dynamics generate a rapid rebound. The top-left panel of Figure 6 shows the pent-up state $x$ rising on impact as suppressed activity accumulates, then gradually declining as agents catch up. By day 1, economic activity is already above baseline as the suppressed activity from day 0 spills forward. The pent-up demand half-life of approximately 1.3 days (implied by $\hat{\delta} = 0.42$) means most of the rebound occurs within the first three days. Despite the rebound, the cumulative effect remains negative: only about half of the impact loss in economic activity is compensated in later

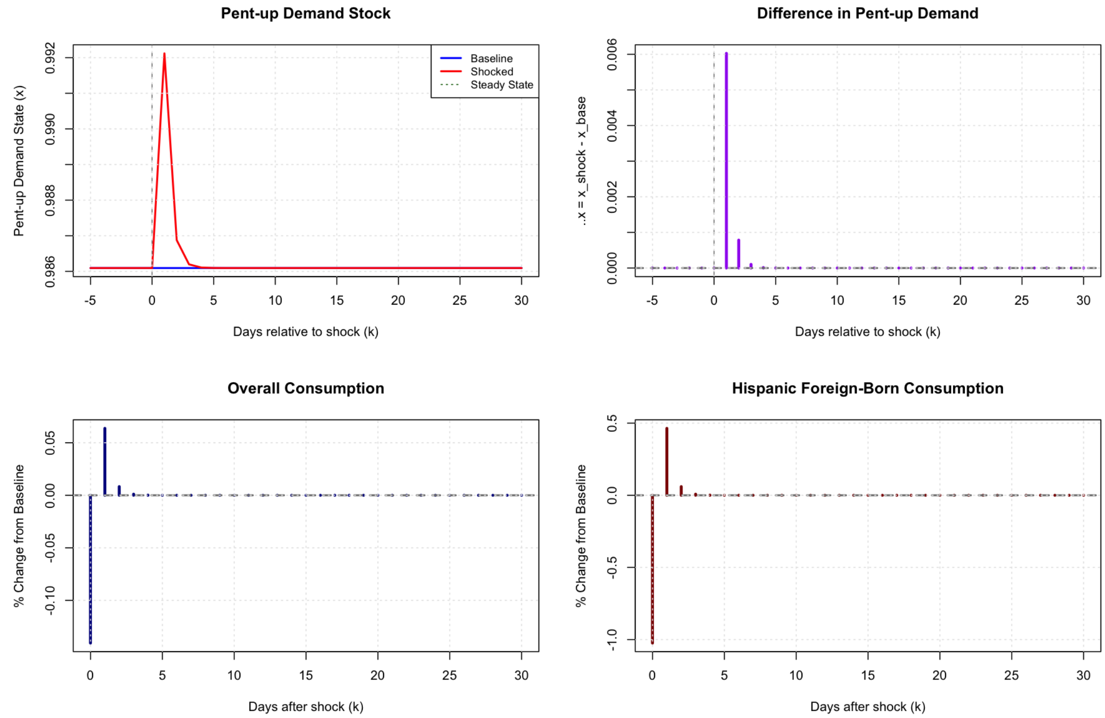



**Figure 6:** Impulse Response to a Belief Shock: Beliefs Revert

*Notes:* Impulse response to a 15 percentage point belief shock from steady state, for the scenario in which beliefs revert to baseline the day after the shock. The figure presents four subfigures: the top left, the evolution of the pent-up demand stock; the top right, the day-to-day changes in the pent-up demand stock; the bottom left, the daily percent changes in economic activity overall; and the bottom right, the daily percent changes in economic activity among the Hispanic foreign born. The companion scenario, in which beliefs update over time after the shock, is shown in Appendix Figure A20.

days. This finding is consistent across both scenarios—whether beliefs revert immediately or evolve endogenously after the shock.

Figure A21 then examines impulse responses across shock magnitudes between 5 and 70 percentage points. The impact and cumulative responses are approximately linear. Notably, the scenario where beliefs evolve endogenously (red lines) shows slightly larger cumulative losses than the immediate-reversion scenario (blue lines), reflecting the persistence of belief shocks when updating is allowed. The structural impulse responses are qualitatively consistent with our reduced-form event studies, which also show an immediate consumption drop followed by recovery. The structural model, however, allows us to distinguish between the short-run and cumulative chilling effect, highlighting that impact effects may overstate the welfare losses when individuals can shift their consumption over time.

## 5.4 Model Validation

We assess the model's performance through goodness-of-fit diagnostics and—most importantly—an exercise built on the structural breaks.

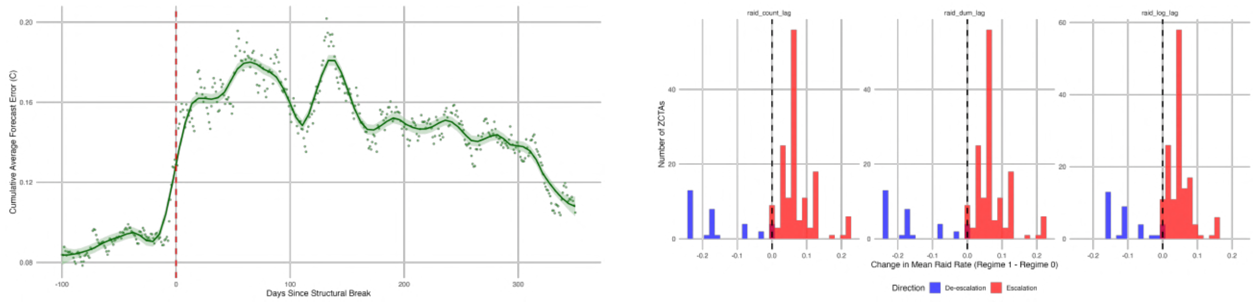
**Model Fit.** Figure A22 presents four fit diagnostics spanning cross-sectional and time-series dimensions. Panel (a) shows a binned scatter plot of predicted versus observed economic activity. The model explains approximately 12 percent of the variation in consumption after removing fixed effects ($R^2 = 0.118$, RMSE = 0.042). Panel (b) shows a binned scatter on the cross-section of ZCTAs, plotting implied mean belief $B_{zd(t)}$ against the observed raid frequencies. Beliefs slightly under-predict raid probabilities at high levels, but the overall correspondence is strong. Panel (c) then shows how model fit varies with the Hispanic foreign-born share. The RMSE is relatively stable across the distribution, indicating that the model does not systematically fit better or worse in communities with different demographic compositions. This is reassuring, since part of our identification relies on cross-sectional variation in the Hispanic share. Finally, panel (d) displays the distribution of model residuals over time. They are well-centered around zero throughout the sample period, albeit with a slight downward drift over time. [20]

**Structural Breaks as Model Validation** Our model implies a specific prediction: when ICE enforcement patterns change unexpectedly, agents' beliefs will temporarily mis-predict raids, causing forecast errors to spike. We test this prediction using the structural breaks identified in Section 4.2 using the Bai and Perron (2003) methodology for the 352 ZCTAs in our estimation sample. We detect 233 structural breaks across 182 ZCTAs (52 percent have at least one break). The median regime duration is 503 days, indicating that enforcement patterns are reasonably stable but do shift periodically.[21]

Panel (a) of Figure 7 displays an event study of mean forecast errors around detected structural breaks. Forecast errors are stable in the 120 days before the break, then spike sharply at event time zero. The increase is substantial—approximately 73 percent above the pre-break level—and highly statistically significant. The spike then gradually decays as agents' beliefs adjust to the new regime, returning to near-baseline levels after approximately

[20] The spikes in the plot, which happen around every 30 days, reveal beginning-of-the-month seasonality in economic activity, which our model did not control for. Within ZCTA, residuals do exhibit some autocorrelation, particularly at short lags as we show in Appendix G.6, Figure A16. This motivates our reporting of HAC standard errors for the model parameter estimates.

[21] Panel (b) of Figure 7 shows the distribution of regime changes by direction. We find that 149 of 182, or 82 percent of detected breaks, represent *escalations* in enforcement intensity—an increase in the underlying raid probabilities—rather than de-escalations. The mean increase is 6.5 percentage points in the daily raid probability.




**(a)** Forecast errors around structural breaks

**(b)** Distribution of regime changes

**Figure 7:** Structural Breaks as Model Validation

*Notes:* Panel (a) displays an event study of mean absolute forecast errors around detected structural breaks, with time normalized to zero at the break date. The shaded region shows the 95 percent confidence interval. Forecast errors spike sharply at the break and gradually decay as beliefs adjust to the new regime. Panel (b) shows the distribution of regime changes by direction: 82 percent of breaks represent escalations in enforcement intensity (red), with a mean increase of 6.5 percentage points in the daily raid probability.

a year. This pattern is what our model predicts: when the world changes in ways agents have trouble anticipating, their prediction-accuracy deteriorates. Because our structural model knows nothing about the structural breaks—which are estimated separately—we find this validation test remarkable.[22]

**Social Cohesion and Learning** Our model estimates a community-level learning process but does not take a stand on the micro-level mechanisms behind it. We expect information diffusion within the local social networks to play a first order role: residents share experiences, warn neighbors, etc. If this is the case, communities with tighter, more interconnected networks should learn more effectively. Our model, which uses no information on such networks, should better capture the relationship between risk perception and economic activity in those ZCTAs.

We test this by regressing the average RMSE of the model residuals on three Facebook-based measures of social cohesion: the clustering coefficient and support ratio from Chetty et al. (2022), which measure how densely interconnected residents' friendship networks are, and the within-ZCTA Social Connectedness Index from Bailey et al. (2018). Table A3 reports the results, controlling for log population and log number of raids. All coefficients are negative, with the clustering coefficient and the support ratio statistically significant. The first principal component of the three measures is also significant. These patterns are consistent with local social networks facilitating the community-level learning we document.

[22] In Appendix H, Figure A23 we present the results of a placebo exercise where we randomly reassign break dates. Placebo breaks show no effect on forecast errors, confirming that the belief dynamics that our model recovers reflect genuine changes in the underlying enforcement patterns.

# 6 Counterfactuals

The structural model provides a natural framework for counterfactual analysis: we simulate economic activity under a series of alternative enforcement scenarios. These simulations allow us to ask questions about the value of information and learning, and to assess the welfare consequences of different enforcement policies.

## 6.1 Methodology

Given our estimated parameters $\hat{\theta} = (\hat{\psi}, \hat{\delta}, \hat{\alpha}_{11}, \hat{\alpha}_{12})$ and the structural shocks $\hat{s}_{zt}$ recovered from the data, we simulate economic activity under alternative enforcement scenarios by solving the model forward. The key feature of this approach is that it captures two sources of endogenous dynamics simultaneously. Pent-up demand evolves as suppressed activity today accumulates into future consumption pressure, and beliefs evolve as agents update their posteriors in response to the enforcement signals they observe.

This realistic treatment of transition dynamics—central to any welfare assessment of policies that generate chilling effects—distinguishes our counterfactuals from comparative-statics exercises that assume instantaneous adjustment. For scenarios involving stochastic elements—such as randomly removing a fraction of observed raids—we use Monte Carlo simulations and report means over the draws.

## 6.2 Raid Likelihood Increase Scenarios

We first consider a policy relevant scenario: what happens if enforcement intensity increases permanently. We simulate scenarios where raid probabilities are scaled up by 25, 50, 75, 100, and 200 percent relative to baseline levels, maintaining each ZCTA's weekday-specific enforcement patterns. Beliefs are initialized at steady-state values matching the scaled probabilities, allowing us to isolate the direct effect of higher enforcement risk on economic activity.

Table 4 summarizes the consumption effects for the Hispanic foreign born across all ZCTAs. The losses grow monotonically with enforcement intensity: a 25 percent rise in raid probability yields a 1.0 percent reduction in economic activity, rising to 2.0 percent at 50 percent, 3.0 percent at 75 percent, and 4.0 percent when raid probabilities are doubled. Appendix Figure A24 reports the full distribution of losses across ZCTAs, which shift leftward and grow more dispersed as enforcement intensifies.

To gauge the magnitude under current enforcement levels, consider the largest scenario—a 200 percent increase, or tripling of raid probabilities. This reduces Hispanic foreign-born economic activity by about 7.5 percent (a median of 8.1 percent across ZCTAs), or

| Scenario | Mean Loss (days/year) | Mean Loss (%) | Median Loss (%) |
|---|---|---|---|
| Baseline (0%) | — | — | — |
| 25% increase | 0.40 | 1.02 | 1.09 |
| 50% increase | 0.79 | 2.02 | 2.16 |
| 75% increase | 1.16 | 3.00 | 3.22 |
| 100% increase | 1.53 | 3.95 | 4.24 |
| 200% increase | 2.93 | 7.55 | 8.13 |

**Table 4:** Consumption Losses Under Increased Enforcement

*Notes:* Consumption losses among the Hispanic foreign born under counterfactual scenarios where raid probabilities are scaled up by k percent. Beliefs are initialized at steady-state values matching the scaled probabilities. Losses are computed relative to the baseline (k=0) and averaged over 50 Monte Carlo simulation draws.

2.9 fewer going-out days per year, from a baseline of roughly 39 going-out days for this high-exposure sample. Using the same national back-of-the-envelope inputs as for the IV estimates—roughly 13.5 million Hispanic foreign-born workers earning about $150 per day—these foregone days amount to on the order of $5.9 billion per year, or about $16 million per day, in foregone earnings nationally.[23]

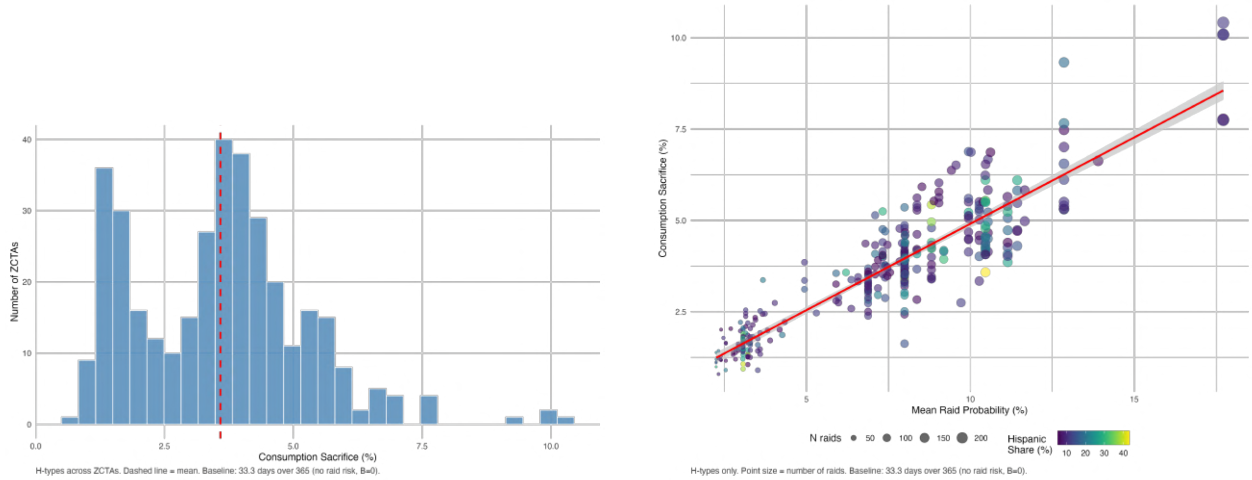
## 6.3 The Value of No Immigration Enforcement

We now ask a different question: how much would people be willing to pay to live in a world with no immigration enforcement at all? To answer this, we compare economic activity under two scenarios. In the first, we set $B_{zd(t)} = 0$ throughout—agents believe no raids will ever occur and go out at their unconstrained optimum. In the second, we set $B_{zd(t)} = \bar{r}_{z,d}$, the average raid probability for ZCTA $z$ on weekday $d$—agents face the true average raid risk with certainty. Crucially, in both scenarios there is no idiosyncratic risk. This ensures an apples-to-apples comparison: we isolate the welfare cost of facing raid risk itself, holding constant the accuracy of beliefs.[24] Under no enforcement ($B = 0$), consumption is higher because agents face no deterrence. Under known risk ($B = \bar{r}$), consumption is lower because agents optimally reduce activity to avoid enforcement exposure. The gap represents the willingness to pay—in consumption terms—for a world without immigration enforcement.

Figure 8 presents the results. Panel (a) shows the distribution of consumption sacrifice

[23]This shares the caveats of the back-of-the-envelope calculation accompanying the IV estimates: we extrapolate from the top-quintile high-exposure sample to the national Hispanic foreign-born workforce, the $150/day figure is approximate, and we count only the extensive margin. Because beliefs are held at their converged steady-state values, it is moreover the fully-anticipated, long-run response, and thus an upper bound on the short-run loss from a comparable surge.

[24]Naturally, an environment without immigration enforcement would likely see changes in patterns of immigration, residential sorting, labor supply, etc. The implications of the exercise here are only partial equilibrium.




**(a)** Distribution of consumption sacrifice **(b)** Consumption sacrifice vs. raid probability

**Figure 8:** Willingness to Pay to Avoid Immigration Enforcement

*Notes:* Panel (a) shows the distribution of consumption sacrifice as a fraction of baseline consumption among the Hispanic foreign born comparing a scenario with no enforcement ($B = 0 \quad \forall t$) to one with known raid risk ($B = \bar{r}_{z,d}$). The mean sacrifice is 3.6 percent of consumption. Panel (b) plots the corresponding consumption changes across ZCTAs against the empirical average raid probabilities. Dot colors denote the Hispanic foreign born share.

among the Hispanic foreign born across the 352 ZCTAs. The mean sacrifice is 3.6 percent of their average consumption (1.2 'staying home' days out of the average 33 'going out' days per year). This represents a substantial and persistent cost: even when agents know the true raid probability with certainty, the mere existence of enforcement risk causes them to reduce economic activity by nearly 4 percent relative to the world where they face none.

Naturally, observed enforcement intensity drives significant heterogeneity in the value of living under no immigration enforcement risk: the scatter plot in panel (b) illustrates that consumption sacrifice rises steeply with the observed average raid probability (correlation coefficient of 0.66). In contrast, the correlation with the Hispanic foreign-born share is essentially zero ($-0.04$). This pattern indicates that enforcement intensity, not demographic composition, determines the welfare costs of living under immigration enforcement.

The 3.6 percent sacrifice we estimate provides a lower bound on the total welfare cost of enforcement, as it captures only the anticipatory behavioral response on the consumption margin, and not the direct welfare costs from being arrested, the psychological costs from living in an environment of persistent uncertainty, or the plethora of other behavioral margins on which people subject to immigration enforcement risk are likely to respond.

In these exercises, agents *know from the start* that enforcement risk is either zero or at its average level; there is no uncertainty about the regime. Thus, there is no learning. If instead we implement an exercise where agents begin with beliefs calibrated to historical enforcement and must learn over time that raids have stopped, we find yearly consumption gains of 1.5 percent relative to the same known-aggregate risk scenario. This is around half the 3.6 percent we estimated under the known no-immigration enforcement world. The ratio of realized gains to potential gains—approximately 42 percent (1.5/3.6)—quantifies how much of the welfare benefit from eliminating enforcement can be captured during a realistic transition period of about a year. The remaining 58 percent represents the cost of learning frictions: even after enforcement stops, it takes time for beliefs to fully adjust.

## 6.4 Perfect Information

We next study the consumption consequences of asymmetric information. To answer this question we use our model estimates to simulate economic activity in a scenario of perfect foresight about immigration enforcement. We suppose agents know whether a raid will occur before making their going-out decision: $B_{zd(t)} = r_{zt}$. This eliminates forecast errors: $\overline{f}_{zt} = 0$ for all $z, t$. Figure A25 presents the results. Panel (a) shows the distribution of consumption changes for H-types across ZCTAs. The mean effect is $-1.7$ percent: under perfect information, economic activity *falls* by about 0.6 days per year (in the data, the average yearly 'going-out' rate is 35 days). This suggests that on average, immigrants underestimate the probability of a raid. Panel (b) shows that these consumption changes vary systematically with forecast accuracy under the baseline. ZCTAs with higher cumulative forecast errors—those whose beliefs were less accurate—reduce economic activity more under perfect information: communities in the lowest forecast-error quartile lower it by only 0.2 percent, against 3.0 percent in the highest quartile. This gradient is a direct consequence of the belief-uncertainty interaction $\alpha_{12}$. In the estimated model the effective sensitivity of activity to beliefs, $\alpha_{11} + \alpha_{12}\overline{f}_{zt}$, is pulled towards zero in communities whose forecasts have been inaccurate because $\hat{\alpha}_{12} = 0.93 > 0$. Perfect information sets $\overline{f}_{zt} = 0$ and restores the full-strength coefficient $\alpha_{11}$, so the chilling response is strongest in communities with the highest in-sample forecast error. In short, the Panel (b) gradient reflects the removal of belief discounting in communities whose forecasts had been least reliable.

## 6.5 Adaptive Learning

We also quantify the implications of discounting noisy signals. We compare economic activity under our estimated model ($\hat{\alpha}_{12} = 0.93$) to a counterfactual where agents are standard Bayesian learners who respond to beliefs with constant intensity ($\alpha_{12} = 0$). Figure A26 presents the results. Panel (a) plots the percent increase in economic activity under our baseline estimates relative to economic activity in the "standard learning" scenario, against the Hispanic foreign-born share. The correlation is essentially zero (0.09), indicating that demographic composition plays almost no role in determining the consumption impacts of discounted learning. Panel (b) shows the distribution of these consumption gains. The mean gain among the Hispanic foreign born is 1.85 percent (0.72 days per year), with gains ranging from near zero to over 5 percent. The variation across ZCTAs is almost entirely driven by forecast error exposure, as expected. Panel (a) illustrates that ZCTAs facing very predictable enforcement patterns (bluest dots in the panel) gain little from belief discounting, while those facing hard to predict patterns (yellowest dots in the panel) gain considerably.

## 6.6 Uniform Enforcement Intensity

Finally, we consider how economic activity would change across ZCTAs if ICE implemented a uniform enforcement intensity policy across them while maintaining its observed enforcement "budget". To do this, we equalize the number of raids per Hispanic foreign born resident across all ZCTAs, so that

$$\rho^* = \frac{R_z}{HFB_z} \qquad \forall z,$$

where $R_z$ denotes the count of raid days, and $HFB_z$ the number of Hispanic foreign born residents in ZCTA $z$. The uniform intensity $\rho^*$ is set to preserve the total observed enforcement budget: $\rho^* = \sum_z R_z / \sum_z HFB_z \approx 22$ raids per 1,000 Hispanic foreign-born residents.[25]

Appendix Figure A27 examines whether this counterfactual represents a meaningful departure from current ICE practice. Panel (a) plots enforcement intensity (raids per Hispanic foreign-born resident) against the Hispanic foreign born share. The correlation is moderately negative ($-0.29$), with most ZCTAs experiencing enforcement intensity rates between 0 and 0.05. Panel (b) shows the redistribution implied by uniform enforcement intensity in terms of average raid probabilities. The more heavily Hispanic foreign born ZCTAs experience the largest increases in raid likelihoods under this policy.

Panel (c) reveals that the implied welfare effects are small. The distribution of consumption gains is tightly concentrated around zero, with a mean of 0.6 percent (about 0.2 days per year). Three-quarters of ZCTAs experience positive gains, but the magnitudes are modest. The small effects reflect that ICE's enforcement intensity is already relatively even, so there is limited scope for uniformity to generate aggregate impacts.

[25] Of course, the composition of the Hispanic foreign born population in terms of lawful status may vary considerably across localities, so this would be an imperfect criterion for uniformity from the perspective of an immigration enforcement agency that cares only about the unlawfully present.

# 7 Conclusions

This paper studies how immigrant communities learn to anticipate and adapt to immigration enforcement. We combine daily bank account transactions data with arrest-level records of ICE operations, and document that enforcement follows predictable weekday patterns and that communities can learn them. Consumption tracks expected raid likelihood, not just realized raids, showing that behavior responds to beliefs about enforcement risk.

We propose a structural model of pent-up demand with Bayesian learning that allows us to decompose chilling effects into consumption that is merely shifted to safer days versus consumption that is permanently foregone. Roughly half of the immediate suppression is recovered through subsequent bounce-back; the other half represents genuine welfare loss. Counterfactual exercises reveal that eliminating enforcement risk would be worth 3.6 percent of Hispanic foreign-born consumption. However, when enforcement is removed, only 42 percent of these potential gains materialize during a realistic transition—the remainder reflects learning frictions as beliefs adjust to the new environment.

Our counterfactuals also speak to the current immigration enforcement environment. We estimate that permanently doubling the likelihood of a raid would, once beliefs have adjusted to the new regime, reduce Hispanic foreign-born consumption by 4%—on the order of $3 billion per year, or about $8.5 million per day, in foregone earnings nationally. Because beliefs adjust only gradually, this steady-state figure is an upper bound on the short-run loss from a comparable enforcement surge. These magnitudes are policy-relevant given the roughly tripling of the yearly number of arrests over 2025-26 compared to the period we study.[26]

Our findings suggest that the benefits of reduced enforcement for households subject to deportation risk depend not only on the magnitude of the reduction but also on how quickly they can learn changes in the enforcement environment. Gradual reductions may generate smaller welfare gains than their steady-state effects would suggest. More broadly, our findings document a form of informational adaptation to state power: populations living under immigration enforcement risk develop forecasting capacity and adjust their economic behavior in response to state action. This adaptation partially mitigates the costs of enforcement uncertainty, but its value is limited by how quickly it allows the detection of policy transitions.

A similar approach can be applied in settings where individuals repeatedly face uncertainty and where event data exhibit predictable structure. Examples include how households

[26]See https://tracreports.org/immigration/detentionstats/book_in_agen_program_table.html. An increase in the number of arrests need not translate one-for-one into a higher per-day likelihood of a raid, so we do not map the threefold rise in arrests directly onto our counterfactual scenarios.

adjust spending or mobility in response to crime risk, how criminals respond to police patrolling intensity, how firms respond to predictable patterns of inspections or regulatory enforcement (including optimal crackdowns), or how individuals adjust mobility, human capital investments (schooling, healthcare utilization) and market participation in response to recurring conflict or violence.

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# Appendices

# A Data Construction

## A.1 Consumption

Data on consumer spending are drawn from the Facteus Debit and Credit Card Panel, a large-scale transaction dataset designed to be representative across income levels, geographic areas, and age groups in the United States. Facteus aggregates anonymized transaction data from a broad set of clients, including major debit, credit, and prepaid card processors, financial institutions, investment firms, and large retail corporations.

The dataset is among the largest proprietary card-transaction panels currently available, covering approximately 40 million unique cardholders, 20 million cards, and $1.3 trillion in consumer spending over the sample period. In aggregate, the panel captures roughly 4 percent of total U.S. consumer spending, which makes it particularly well suited for analysis at fine geographic resolutions, such as ZIP codes. The data were made available to us through a partnership between the Federal Reserve Bank of Chicago and Facteus.

The raw data are observed at the transaction level and include an anonymized account identifier, transaction date, ZIP code of residence of the account holder, transaction type (spend, deposit, ATM withdrawal, or load), and transaction value. We retain only accounts that are active both at the beginning and at the end of the observation window, defined as having at least one recorded transaction in both 2014 and 2018. This restriction mitigates bias arising from account entry and exit into the panel.

For the analysis, we aggregate transactions up to the ZCTA–day level. To ensure adequate and stable coverage across locations, we impose a "selection cone" that requires a minimum of 30 active accounts per ZIP code and a penetration rate between 3 and 25 accounts per 1,000 residents. This procedure excludes areas with insufficient data coverage as well as ZIP codes with implausibly high representation in the panel.

## A.2 Geographic Mapping

The TRAC arrest records identify enforcement events at the county level, while our financial transaction data and demographic variables are measured at the ZIP Code Tabulation Area (ZCTA) level. Since ZCTAs do not nest cleanly within counties—many ZCTAs straddle county boundaries—we develop a two-step mapping procedure to assign county-level arrests to ZCTAs.

The first step maps county names in the TRAC data to county FIPS codes. The TRAC records identify counties by name and state, but do not include standardized FIPS codes. We use a county crosswalk file developed for Ciancio and García-Jimeno (2024), which maps county names to FIPS codes and was constructed to handle variations in county naming conventions across data sources.

The second step maps counties to ZCTAs. We use the Census Bureau's ZCTA-to-county relationship file, which reports the share of each ZCTA's population residing in each overlapping county. We associate a ZCTA with a county if either (i) more than 2.5 percent of the ZCTA's population resides in that county, or (ii) the county is the only one the ZCTA overlaps with. Under this rule, ZCTAs may be associated with multiple counties—in our data, up to five.

For each ZCTA-date observation, we sum arrests across all associated counties. This approach treats a ZCTA as "exposed" to enforcement in any of its associated counties. For example, if ZCTA $z$ is associated with counties $c_1$ and $c_2$, then arrests in ZCTA $z$ on date $t$ equals the sum of arrests in $c_1$ and $c_2$ on date $t$. This assignment rule implies that the same county-level arrest may contribute to the arrest count for multiple ZCTAs, which is appropriate given that residents of border-straddling ZCTAs may work, shop, or travel in any of their associated counties.

The mapping procedure yields 17,765 ZCTAs with at least one associated county appearing in the TRAC arrest records. After merging with Facteus transaction data and ACS demographics and applying sample restrictions (see Section A.3), the final analysis sample contains 12,865 ZCTAs.

## A.3 Sample Selection

The analysis sample is constructed by merging three data sources: financial transaction data from Facteus, ICE interior arrest records from TRAC, and demographic characteristics from the 2018 American Community Survey 5-Year estimates. Table A1 documents the sample restrictions applied and their effects on sample size.

| Step | Restriction | ZCTAs | ZCTA-Days |
|---|---|---|---|
| 0 | Facteus data (all zip codes) | 39,648 | 53,088,672 |
| 1 | Merge with Raids data | 39,648 | 53,088,672 |
| 2 | Merge with ACS data | 30,177 | 40,407,003 |
| 3 | Selection cone: accts > 30, accts.pc ∈ [3, 25) | 12,865 | 17,226,235 |

**Table A1:** Sample Selection

*Notes:* Sample construction from Facteus financial transaction data, TRAC ICE arrest records, and 2018 ACS 5-Year estimates. ZCTA-Days computed as ZCTAs × 1,339 days (October 2014 – May 2018).

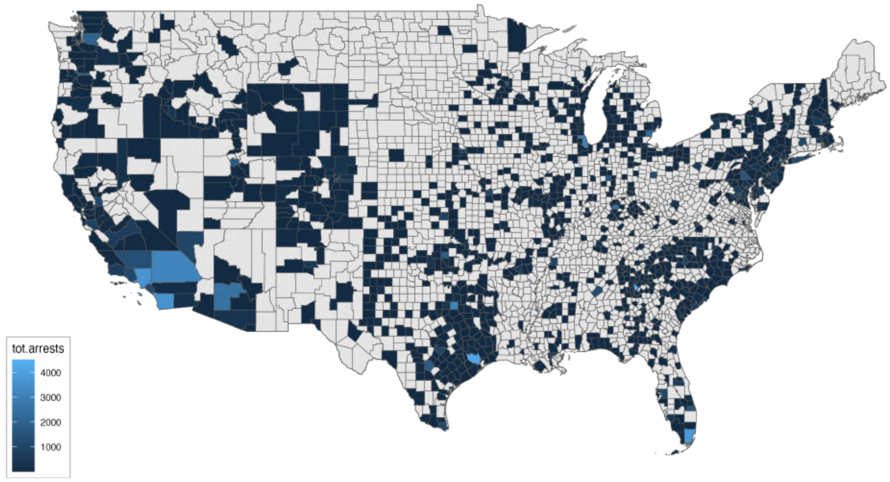
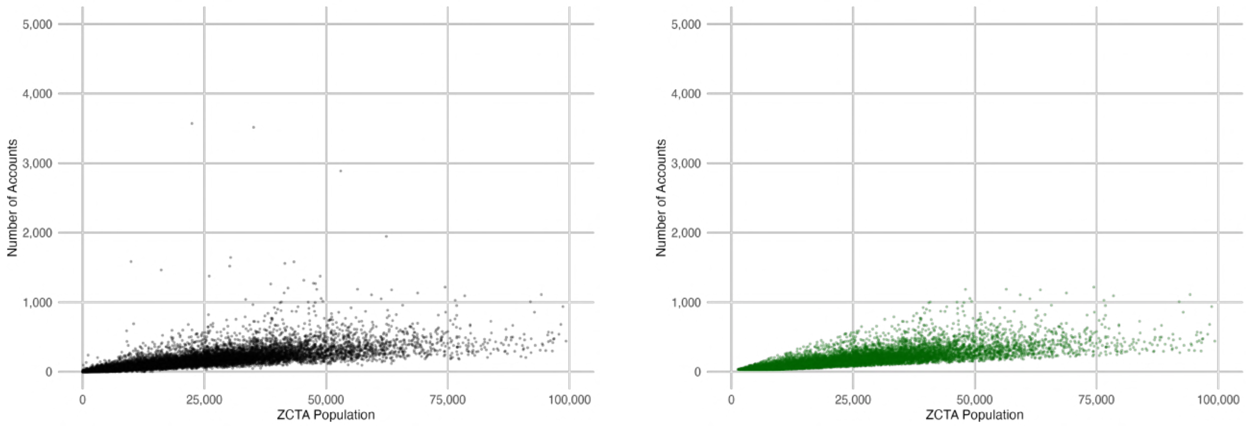



**Figure A1:** Geographic Distribution of ICE Interior Arrests, October 2014 – May 2018.
*Notes:* Total ICE interior raid arrests by county over the sample period. Brighter shading indicates higher arrest counts.

The starting point is the universe of ZIP codes appearing in the Facteus data during the sample period (October 2014 through May 2018). We first merge ICE arrest records, assigning zero arrests to ZIP codes with no recorded enforcement activity. We then perform an inner join with ACS demographic data, restricting to ZIP Code Tabulation Areas (ZCTAs) that appear in both datasets. This reduces the sample from 39,648 to 30,177 ZCTAs, reflecting the imperfect correspondence between commercial ZIP codes and Census-defined ZCTAs.

We impose one additional sample restriction: a "selection cone" requiring ZCTAs to have more than 30 accounts, at least 3 accounts per 1,000 population, and fewer than 25 accounts per 1,000 population. This ensures sufficient account coverage for meaningful activity measures while excluding ZCTAs with implausibly high or low penetration rates. Figure A2 illustrates the selection cone graphically.

The final sample contains 12,865 ZCTAs observed over 1,339 days, for a total of 17,226,235 ZCTA-day observations.




**(a)** All ZCTAs **(b)** Selection cone applied

**Figure A2:** Selection Cone for Sample Restriction

*Notes:* Scatter plots of ZCTA population against number of Facteus accounts. Left panel shows all ZCTAs after merging with ACS data. Right panel shows ZCTAs satisfying the selection cone restriction: more than 30 accounts, at least 3 accounts per 1,000 population, and fewer than 25 accounts per 1,000 population.

# B Operation Crosscheck

Figure A3 displays the geographic distribution of ZCTAs included in the Operation Crosscheck event study sample. It includes 802 ZCTAs that experienced at least one arrest during the Crosscheck week (March 1–5, 2015) and had no arrests in the 10 days before or after.

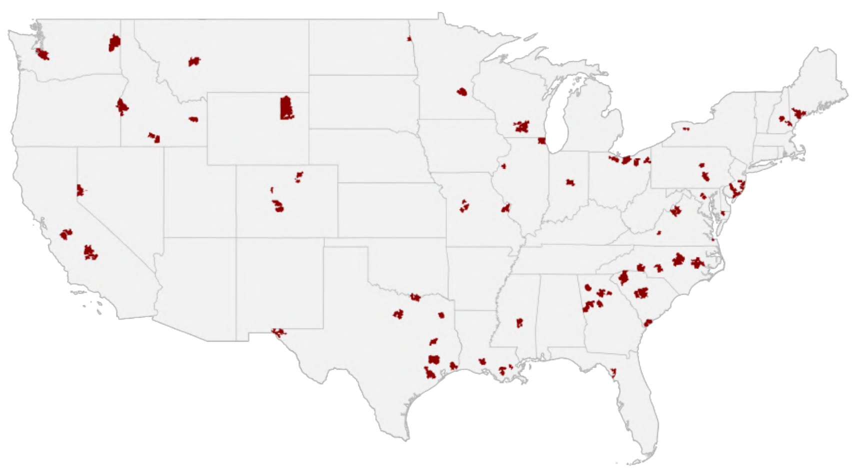


**Figure A3:** Geographic Distribution of Operation Crosscheck Event Study Sample

*Notes:* Map shows the 802 ZCTAs included in the Crosscheck event study sample. ZCTAs experienced at least one arrest during Operation Crosscheck (March 1–5, 2015) with no arrests in the 10 days before or after.

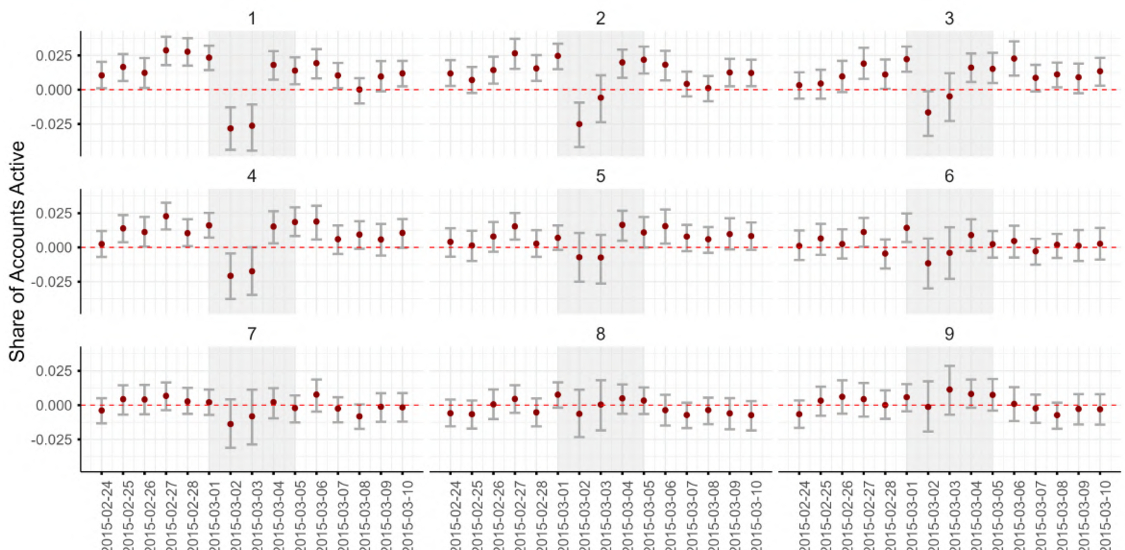



**Figure A4:** Operation Crosscheck Event Study: Effects by Hispanic Foreign-Born Decile

*Notes:* Event study coefficients by Hispanic foreign-born share decile. Decile 1 corresponds to ZCTAs with the highest Hispanic foreign-born shares; Decile 10 corresponds to the lowest. Sample and estimation as in Figure 2. Error bars show 95 percent confidence intervals from 1,000 Bayesian bootstrap replications (Rubin, 1981) clustered at the ZCTA level.

# C Enforcement Patterns

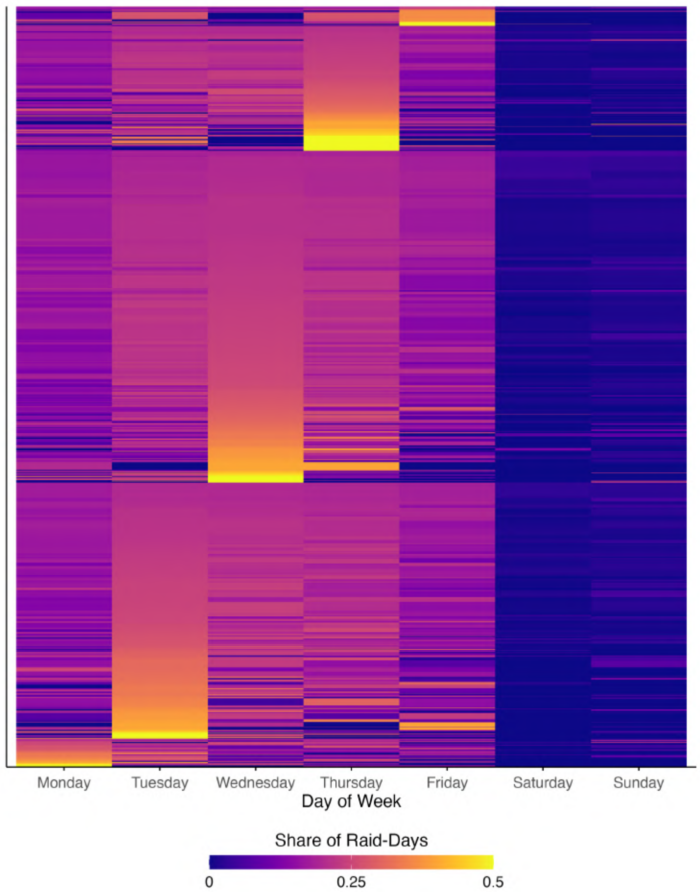



**Figure A5:** Heterogeneity in Weekday Enforcement Patterns Across ZCTAs

*Notes:* Heatmap of weekday enforcement profiles for 6,711 ZCTAs with raids between 2014 and 2018. Each row represents one ZCTA. Each column shows the share of that ZCTA's raid-days occurring on each weekday. ZCTAs are grouped by peak enforcement day and sorted by concentration within each group. While mid-week enforcement is generally higher (vertical structure), the intensity varies substantially across ZCTAs (horizontal variation within each block).

# D Event Studies Robustness

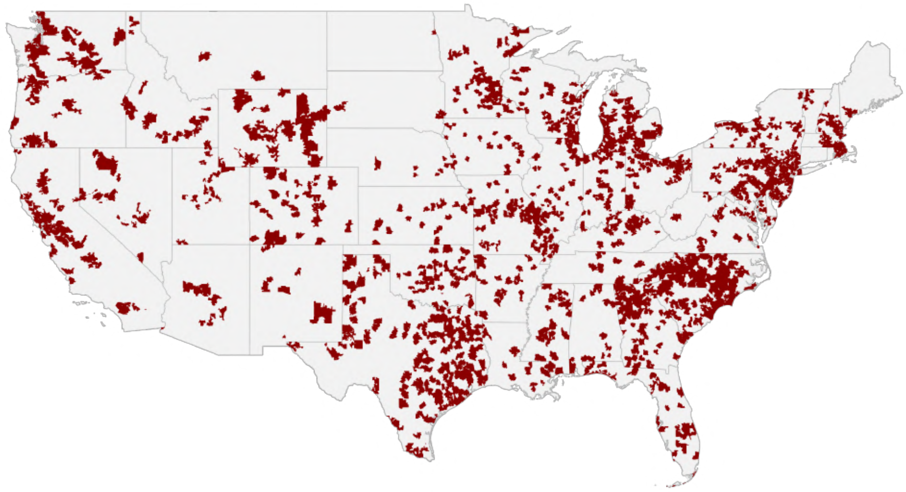


**Figure A6:** Geographic Distribution of Event Study Sample

*Notes:* Map shows the 5,967 ZCTAs included in the event study second stage. These ZCTAs experienced at least one "clean" raid event (no other raids within ±8 days) during the sample period.

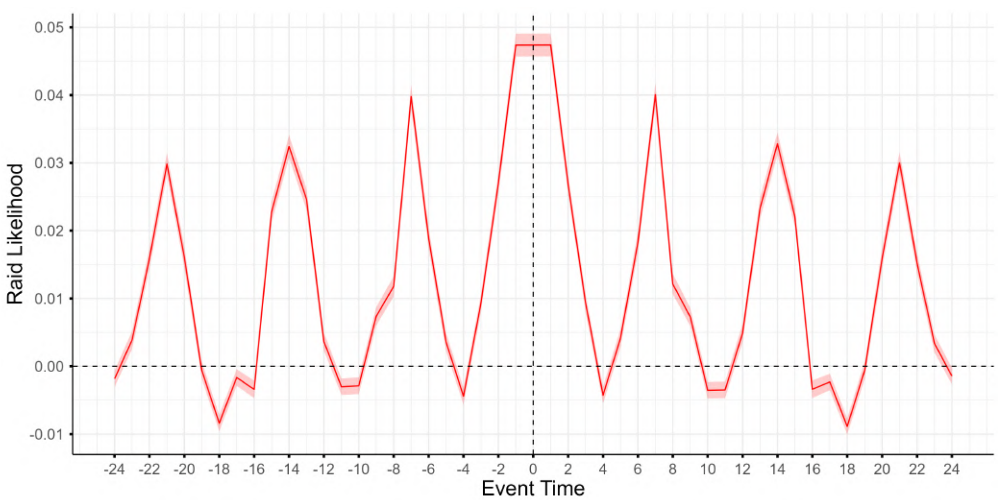



**Figure A7:** Raid Likelihood from Autocorrelation Regression

*Notes:* Coefficients from regressing the raid indicator on 24 leads and lags of itself, with ZCTA and date fixed effects. The seven-day periodicity reflects ICE's tendency to conduct raids on preferred weekdays. Sample restricted to ZCTAs in the event study estimation sample.

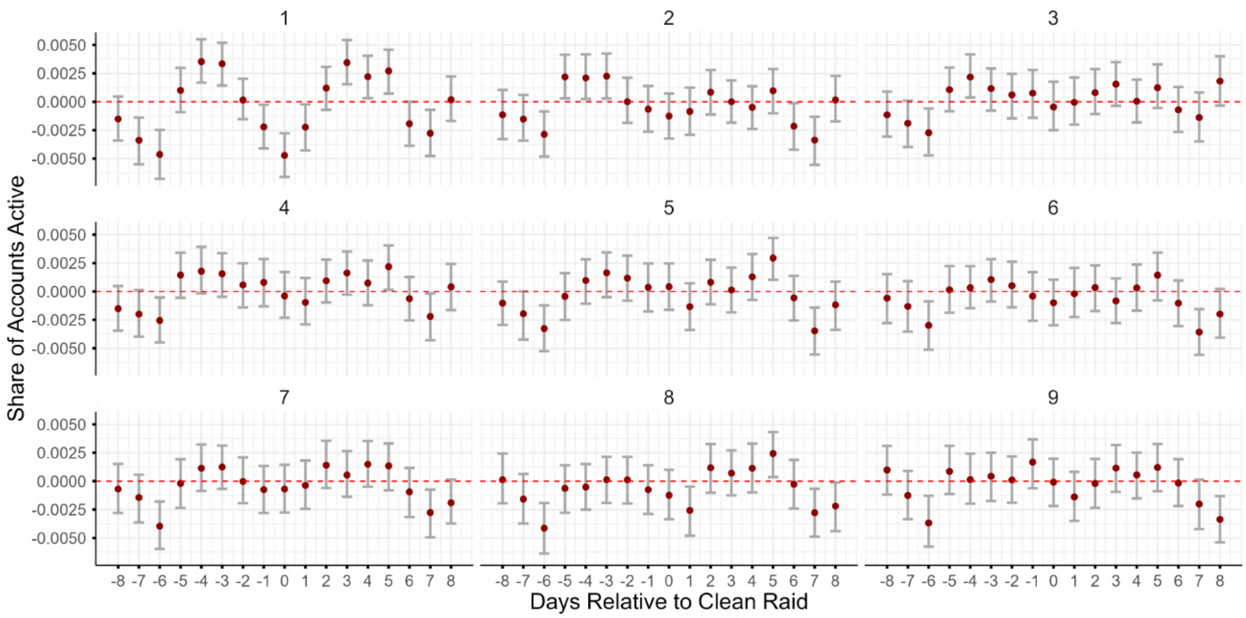



**Figure A8:** Clean Raids Event Study: Effects by Hispanic Foreign-Born Share Decile

*Notes:* Event study coefficients by decile of Hispanic foreign-born share. Decile 1 corresponds to ZCTAs with the highest Hispanic foreign-born shares; Decile 9 corresponds to the lowest shown. The seven-day cyclical pattern is strongest and most statistically significant in the top deciles. Effects attenuate progressively toward lower deciles. Sample includes 33,091 clean raid events across 5,967 ZCTAs. Error bars show 95 percent confidence intervals from 1,000 Bayesian bootstrap replications (Rubin, 1981) clustered at the ZCTA level.

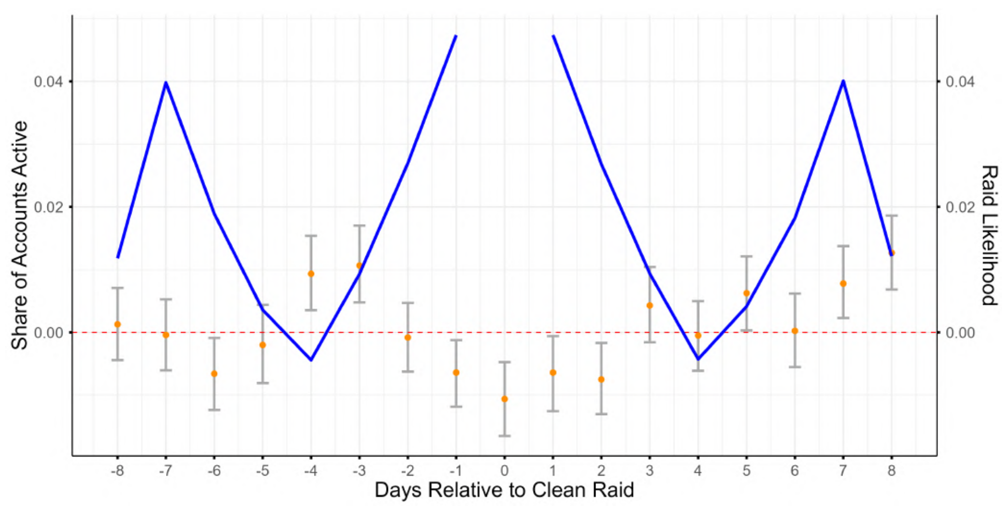



**Figure A9:** Clean Raids Event Study: Hispanic Foreign-born Share Interactions including ZCTA $\times$ Weekday Fixed Effects

*Notes:* Consumption coefficients (orange, left axis) overlaid with raid likelihood (blue, right axis) by days relative to clean raid. The event-study specification includes ZCTA $\times$ weekday and date fixed effects. The event study coefficients are those on the interaction between event time and the Hispanic foreign-born share. Error bars show 95 percent confidence intervals from 1,000 Bayesian bootstrap replications (Rubin, 1981) clustered at the ZCTA level.

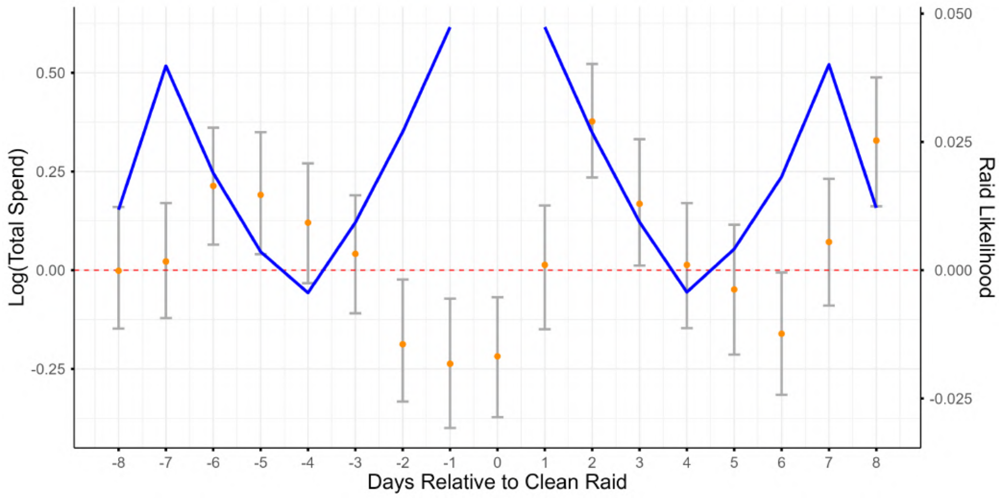



**Figure A10:** Event Study Overlay: Log Total Spending

*Notes:* Consumption coefficients (orange, left axis) overlaid with raid likelihood (blue, right axis) by days relative to clean raid. Outcome variable is log total spending. The event study coefficients are those on the interaction between event time and the Hispanic foreign-born share. Error bars show 95 percent confidence intervals from 1,000 Bayesian bootstrap replications (Rubin, 1981) clustered at the ZCTA level.

# E Measurement Error Derivations

This appendix provides the full derivation of the OLS bias result and sign condition stated in Section 3.4.

## E.1 Setup and Notation

Let $r_{zt} \in \{0, 1\}$ denote whether a raid occurs in ZCTA $z$ on date $t$, and let $r^e_{zt} \equiv \mathbb{E}[r_{zt}|\mathcal{I}_{zt}]$ denote agents' subjective belief about raid likelihood given their information set. Define the prediction error as $\tilde{\eta}_{zt} \equiv r_{zt} - r^e_{zt}$. Let $\varphi^* \equiv \mathbb{P}(r_{zt} = 1)$ denote the true (unconditional) raid probability.

The model is

$$y_{zt} = \alpha_z + \delta_t + \beta r^e_{zt} + \epsilon_{zt}$$

Using $r_{zt}$ as a proxy for $r^e_{zt}$ yields

$$y_{zt} = \alpha_z + \delta_t + \beta r_{zt} + (\epsilon_{zt} - \beta \tilde{\eta}_{zt})$$

Assuming $\text{cov}(r_{zt}, \epsilon_{zt}) = 0$ conditional on fixed effects, the OLS estimand is

$$\beta_{OLS} = \beta + \frac{\text{cov}(r_{zt}, \epsilon_{zt} - \beta\tilde{\eta}_{zt})}{\text{var}(r_{zt})} = \beta \left[1 - \frac{\text{cov}(r_{zt}, \tilde{\eta}_{zt})}{\text{var}(r_{zt})}\right]$$

We now compute $\text{cov}(r_{zt}, \tilde{\eta}_{zt})$. First:

$$\begin{aligned}
\mathbb{E}[r_{zt}\tilde{\eta}_{zt}] &= \mathbb{E}\left[\mathbb{E}[r_{zt}\tilde{\eta}_{zt}|r_{zt}]\right] \\
&= \mathbb{P}(r_{zt}=1)\mathbb{E}[\tilde{\eta}_{zt}|r_{zt}=1] \\
&= \varphi^*\mathbb{E}[r_{zt}-r^e_{zt}|r_{zt}=1] \\
&= \varphi^*(1-\mathbb{E}[r^e_{zt}|r_{zt}=1])
\end{aligned}$$

Next:

$$\begin{aligned}
\mathbb{E}[\tilde{\eta}_{zt}] &= \mathbb{E}\left[\mathbb{E}[\tilde{\eta}_{zt}|r_{zt}]\right] \\
&= \varphi^*\mathbb{E}[r_{zt}-r^e_{zt}|r_{zt}=1] + (1-\varphi^*)\mathbb{E}[r_{zt}-r^e_{zt}|r_{zt}=0] \\
&= \varphi^*(1-\mathbb{E}[r^e_{zt}|r_{zt}=1]) - (1-\varphi^*)(\mathbb{E}[r^e_{zt}|r_{zt}=0])
\end{aligned}$$

Therefore:

$$\begin{aligned}
\text{cov}(r_{zt}, \tilde{\eta}_{zt}) &= \mathbb{E}[r_{zt}\tilde{\eta}_{zt}] - \mathbb{E}[r_{zt}]\mathbb{E}[\tilde{\eta}_{zt}] \\
&= \varphi^*(1-\mathbb{E}[r^e_{zt}|r_{zt}=1]) \\
&\quad - \varphi^*\left\{\varphi^*(1-\mathbb{E}[r^e_{zt}|r_{zt}=1]) - (1-\varphi^*)(\mathbb{E}[r^e_{zt}|r_{zt}=0])\right\} \\
&= \varphi^*(1-\varphi^*)\left\{1-\mathbb{E}[r^e_{zt}|r_{zt}=1] + \mathbb{E}[r^e_{zt}|r_{zt}=0]\right\}
\end{aligned}$$

Since $\text{var}(r_{zt}) = \varphi^*(1-\varphi^*)$ for a Bernoulli random variable:

$$\begin{aligned}
\beta_{OLS} &= \beta\left[1 - \frac{\varphi^*(1-\varphi^*)\left\{1-\mathbb{E}[r^e_{zt}|r_{zt}=1] + \mathbb{E}[r^e_{zt}|r_{zt}=0]\right\}}{\varphi^*(1-\varphi^*)}\right] \\
&= \beta\left[1 - \left\{1-\mathbb{E}[r^e_{zt}|r_{zt}=1] + \mathbb{E}[r^e_{zt}|r_{zt}=0]\right\}\right] \\
&= \beta\left[\mathbb{E}[r^e_{zt}|r_{zt}=1] - \mathbb{E}[r^e_{zt}|r_{zt}=0]\right]
\end{aligned}$$

This is the OLS bias result stated in equation (8).

## E.2 Condition for Correct Sign

When does the OLS estimand have the same sign as the true $\beta$? Using Bayes' rule, we can express the conditional expectations as:

$$\mathbb{E}[r^e_{zt}|r_{zt}=r] = \int r^e_{zt}\frac{\mathbb{P}(r_{zt}=r|r^e_{zt})f(r^e_{zt})}{\mathbb{P}(r_{zt}=r)}dr^e_{zt}$$

Let $P(r^e_{zt}) \equiv \mathbb{P}(r_{zt}=1|r^e_{zt})$ denote the true raid probability given beliefs. After some algebra (expanding and simplifying), the difference $\mathbb{E}[r^e_{zt}|r_{zt} = 1] - \mathbb{E}[r^e_{zt}|r_{zt} = 0]$ has the same sign as $\beta$ if and only if:

$$\text{cov}(r^e_{zt}, P(r^e_{zt})) \geq 0$$

This condition says that the OLS has the correct sign as long as beliefs and true raid probabilities are positively correlated: when agents believe raids are more likely, actual raid probabilities should also be higher. This condition is satisfied whenever learning is taking place, since learning implies that high beliefs correspond to genuinely high-risk periods. Notice that when beliefs contain no information about raid likelihood (i.e., $P(r^e_{zt}) = \varphi^*$ for all $r^e_{zt}$), then $\text{cov}(r^e_{zt}, P(r^e_{zt})) = 0$ and the OLS estimand is zero. Thus, OLS can be interpreted as a test of whether beliefs contain information about the raid process.

# F Regime Changes

How stable are ICE's enforcement patterns? If the agency occasionally changes strategy, communities face a more complex learning problem. We document that despite overall predictability, ICE does change patterns at specific times and places, creating regime changes that communities must adapt to.

### F.0.1 Detecting Regime Changes

We apply the Bai and Perron (2003) structural break detection method to weekday-specific raid probabilities, testing for significant changes in these probabilities within ZCTAs. The method estimates break dates by finding the partition of the sample that globally minimizes the sum of squared residuals, using an efficient dynamic programming algorithm.[27]

For each ZCTA, we estimate

$$r_{z,t} = \sum_{d=\text{Mon}}^{\text{Sun}} \beta^{(j)}_d \cdot \mathbf{1}[d(t)] + \gamma^j \cdot r_{z,t-1} + \epsilon_{z,t} \tag{19}$$

where $j = 0, 1, ...$ denote the underlying regimes to be detected, $r_{z,t}$ is a raid indicator (or count), $\mathbf{1}[d(t)]$ are weekday dummies, and $r_{z,t-1}$ captures serial correlation. The structural

[27]Since partitioning the sample will always weakly reduce the SSR even absent true breaks, a formal statistical test is required. The test compares the minimized SSR under the alternative of $k$ breaks against the null of no breaks, and uses critical values that account for the choice of a maximal test statistic. To determine the number of breaks, the method evaluates whether adding an additional break significantly improves fit over a model with fewer breaks. This test requires fixing the minimum segment (regime) length as a parameter. We fix it to 15 percent of the sample (200 days), allowing for up to 6 breaks.

break algorithm identifies dates where the weekday coefficients $\{\beta_d\}$ shift significantly.[28]

To ensure robustness, we estimate breaks using three specifications for the dependent variable: a raid dummy, raid counts, and log raid counts. We grade detected breaks based on agreement across specifications: a break receives an "A" grade if all three specifications identify a break within 7 days of each other; a "B" grade if they agree within 28 days; and a "C" grade if at least two specifications agree within 14 days. We restrict this analysis to ZCTAs in the top quintile of the Hispanic foreign-born share —the communities we identified in our event study exercises as responding to immigration enforcement risk—, with at least 191 raid days —with sufficient enforcement activity to reliably detect pattern changes.

### F.0.2 Prevalence and Timing of Regime Changes

Figure A11 displays detected break dates. We identify 513 breaks in 439 ZCTAs, and no breaks for the remaining ZCTAs. The distribution spikes in January 2017 (around President Trump's first inauguration), but the mean break date is August 31, 2016—regime changes occurred throughout the sample.

Figure A12 presents break timing by ICE district. Some districts (Boston, Los Angeles) show tight clustering around January 2017; others (San Francisco, San Antonio) are more dispersed. Break dates cluster within districts, suggesting enforcement shifts occur largely at the AOR level.

### F.0.3 Nature of Regime Changes: Intensity vs. Re-shuffling

An important question is whether regime changes reflect shifts in overall enforcement *intensity* (more or fewer raids on all days) or *re-shuffling* of weekday patterns (different days become high- vs. low-activity days). The distinction matters for learning: intensity shifts are easier to detect and adapt to, while re-shuffling of patterns requires relearning which days are risky.

To assess this, we compute the correlation between weekday coefficients across regimes for each ZCTA with a detected break. If a break reflects a pure intensity shift, the weekday coefficients in the new regime should be highly correlated with those in the old regime (the ranking of days by risk is preserved). If instead the break reflects re-shuffling, the correlation should be low or negative.

[28] The Bai and Perron (2003) approach allows a 'partial' regime change model where coefficients on covariates other than the main ones of interest remain constant across regimes, or a 'pure' regime change model where they are also allowed to vary. In our case, we implement the 'pure' model, allowing the coefficient on the lagged raid variable, $\gamma^j$, to also vary across regimes.

Figures A13a and A13b display these correlations for Trump-window breaks (within one month of the inauguration) and all other breaks. Both distributions center on large positive values: regime changes primarily reflect intensity shifts, not re-shuffling. The Trump transition intensified enforcement but did not alter which days were high or low risk. ICE does not appear to shuffle weekday patterns to maintain unpredictability.

### F.0.4 Implications for Learning

The presence of regime changes has implications for how communities form beliefs about enforcement risk. If enforcement patterns were perfectly stable, a simple decreasing-gain learning rule (e.g. a Kalman filter) would eventually converge to the true raid probabilities. In such a context, weighting all past observations equally is informationally optimal, and beliefs become increasingly precise over time.

But with regime changes, standard Bayesian updating would fail to converge to the underlying enforcement rates. When ICE changes its patterns, old observations become misleading—they reflect a stale enforcement regime rather than current policy. Agents face a trade-off: weighting recent observations more heavily allows faster adaptation to new patterns but increases noise. This trade-off motivates the learning model specification we will introduce below, where the effective learning rate responds to forecast errors.

The finding that regime changes primarily involve intensity shifts rather than re-shuffling has a silver lining: while agents must update their beliefs about overall risk levels, they can continue to rely on previously learned weekday patterns. This reduces the learning burden and we believe it helps explain why communities can maintain belief-tracking behavior even in an environment with occasional policy changes.

Figure A11 presents the distribution of structural break dates across all ZCTAs with breaks.

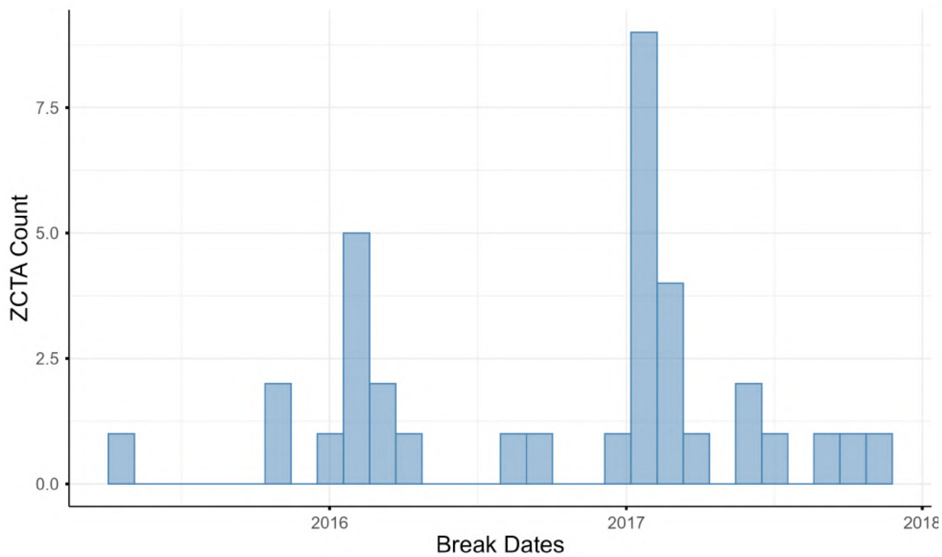



**Figure A11:** Distribution of Detected Structural Breaks in ICE Enforcement

*Notes:* Histogram of structural break dates detected using Bai and Perron (2003) methodology. 513 breaks across 439 ZCTAs. Sample restricted to top-quintile Hispanic foreign-born ZCTAs with at least 191 raid days. Breaks graded by agreement across three specifications (raid dummy, raid count, log raid count with weekday dummies and lag).

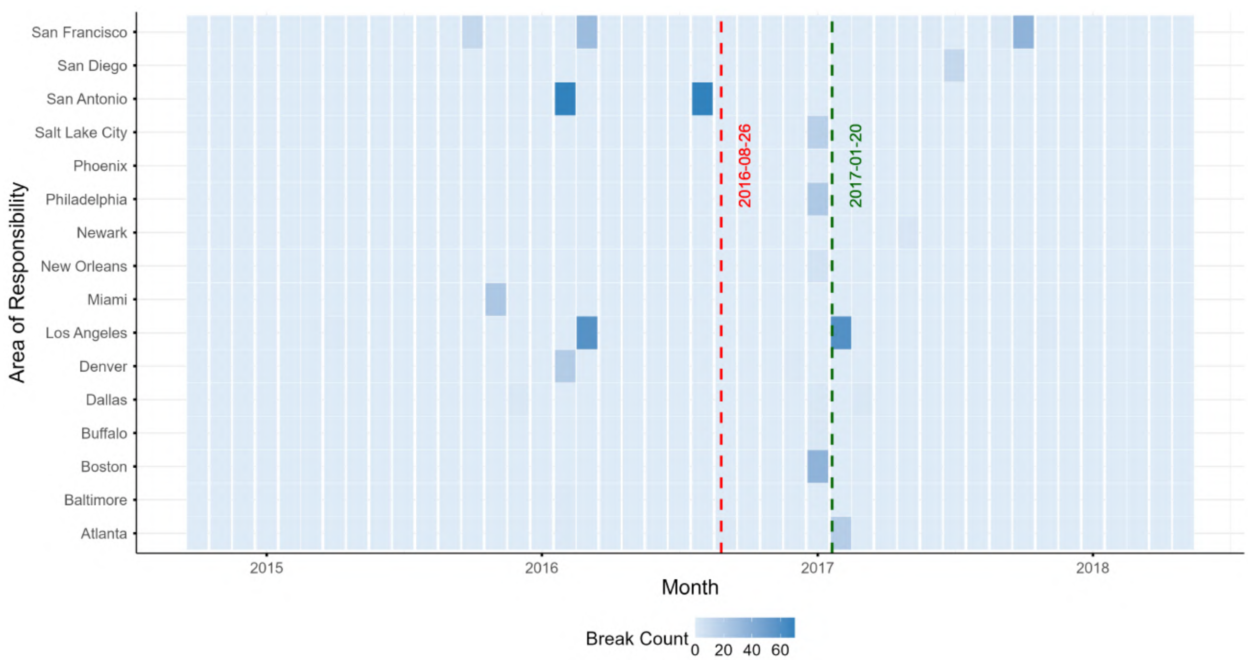
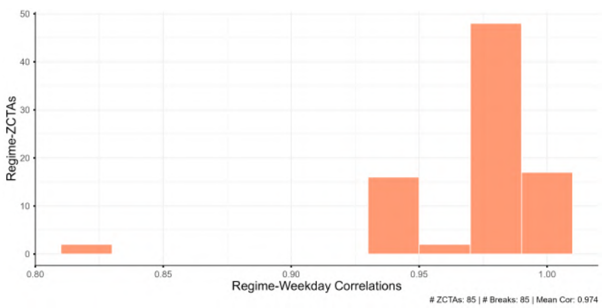
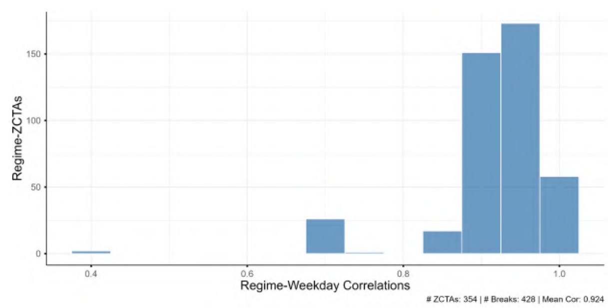



**Figure A12:** Timing of Structural Breaks by ICE District

*Notes:* Distribution of structural break dates by ICE Area of Responsibility (district). Structural break dates detected using Bai and Perron (2003) methodology. Sample restricted to top-quintile Hispanic foreign-born ZCTAs with at least 191 raid days. Breaks graded by agreement across three specifications (raid dummy, raid count, log raid count with weekday dummies and lag). Districts ordered by median break date. Red dashed line indicates the overall median break date. Green dashed line indicates President Trump's first inauguration. N = 513 breaks across 439 ZCTAs.




**(a)** Breaks within a month of Trump Inauguration




**(b)** Breaks outside a month of Trump Inauguration

**Figure A13:** Correlation of Weekday Betas Across Regimes

*Notes:* Distribution of correlations between weekday coefficients in regime 0 (before break) and regime 1 (after break). Structural break dates detected using Bai and Perron (2003) methodology. Sample restricted to top-quintile Hispanic foreign-born ZCTAs with at least 191 raid days. Breaks graded by agreement across three specifications (raid dummy, raid count, log raid count with weekday dummies and lag). Panel (a) includes ZCTAs with detected breaks within one month of Trump inauguration (N = 85 ZCTAs); panel (b) includes breaks more than one month from inauguration (N = 354 ZCTAs). High correlation indicates pure intensity shift (same days remain high/low risk); low correlation indicates re-shuffling of weekday patterns.

# G Estimation Details

This appendix provides technical details on the estimation procedure summarized in Section 5.

## G.1 Value Function Solution

For a given parameter vector $\boldsymbol{\theta} = (\psi, \delta, \alpha_{11}, \alpha_{12})$, we solve for the stationary value function $V(x)$ by value iteration over a discrete grid.

**Grid construction.** We discretize the pent-up state space $x \in [x_{\min}, x_{\max}]$ using a grid of 401 equally-spaced points. The upper bound $x_{\max} = 25$ ensures the stationary distribution of $x$ has negligible mass near the boundary.

**Iteration.** Starting from an initial guess $V^{\tau(0)}(x) = 0$, we iterate on the Bellman equation

$$V^{\tau(k+1)}(x) = \log\Big(\exp\big(c(x) + \beta V^{\tau(k)}((1-\delta)x + \kappa)\big) + \exp\big(c(x) + \beta V^{\tau(k)}((1-\delta)x)\big)\Big), \quad (20)$$

where $c(x) = -(\psi/2)x^2$ is the flow cost. We compute between-grid values by linear interpolation. The iteration continues until $\|V^{\tau(k+1)} - V^{\tau(k)}\| < 10^{-8}$.

**Dynamic incentive.** Once convergence is achieved, we compute the dynamic incentive term

$$\Delta(x)^{\tau} \equiv \beta\big[V^{\tau}((1-\delta)x) - V^{\tau}((1-\delta)x + \kappa)\big]. \quad (21)$$

This quantity enters the probabilities and captures the option value of going out today versus accumulating additional pent-up demand.

Figure A14 displays the estimated value function and dynamic incentive term at the estimated parameter values.

## G.2 Belief Construction

**Prior initialization.** We estimate the prior from a probit regression on the first 105 days (burn-in period):

$$r_{zt} = \Phi\big(\gamma_{d(t),r_{t-1}} + \beta^{h}_{d(t),r_{t-1}} h_z + \beta^{y}_{d(t),r_{t-1}} y_z + \eta_{\mathrm{AOR}(z)}\big), \quad (22)$$

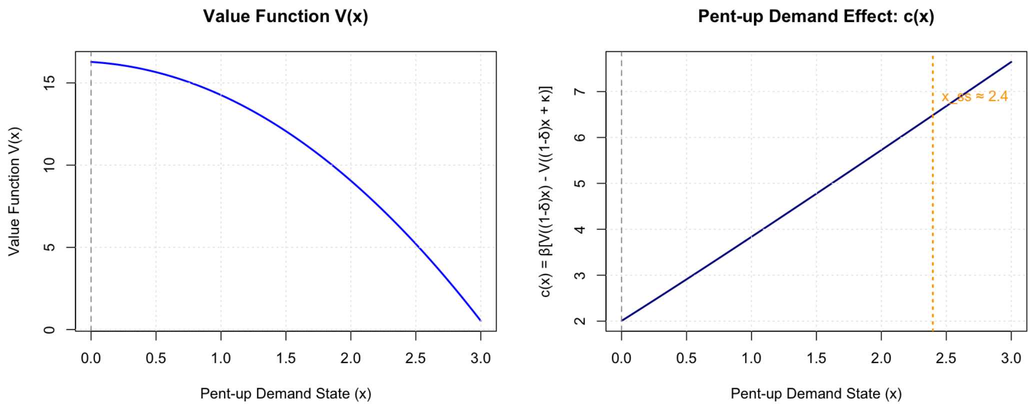



**Figure A14:** Estimated Value Function and Dynamic Incentive

*Notes:* Left panel shows the value function $V(x)$ as a function of the pent-up demand state $x$. Right panel shows the dynamic incentive term $\Delta(x) = \beta[V((1-\delta)x) - V((1-\delta)x + \kappa)]$, which captures the option value of going out today versus accumulating additional pent-up demand. Both computed at the estimated parameter values $\hat{\psi} = 2.23$ and $\hat{\delta} = 0.42$.

where $\gamma_{d(t),r_{t-1}}$ are weekday $\times$ previous-day-raid intercepts, $\beta^h_{d(t),r_{t-1}}$ and $\beta^y_{d(t),r_{t-1}}$ are weekday $\times$ previous-day-raid slopes on Hispanic foreign-born share $h_z$ and log median household income $y_z$, respectively, and $\eta_{\text{AOR}(z)}$ are ICE Area of Responsibility fixed effects. The predicted probability $\hat{q}_{z,d,k}$ initializes the Beta prior mean, with $a^0 + b^0 = 10$. Table A2 reports the probit estimates.

**Bayesian updating.** For each ZCTA $z$, weekday $d$, and lag state $k \in \{0, 1\}$, we compute the Beta parameter updates according to equation (16).

**Cumulative forecast error.** We compute the running average absolute forecast error

$$\bar{f}_{zt} = \frac{1}{t - t_0} \sum_{s=t_0}^{t-1} \omega_s |r_{zs} - B_{zs}|, \tag{23}$$

where $t_0$ is the end of the burn-in period. This measure enters the heterogeneous belief coefficient $\alpha_1(\bar{f}) = \alpha_{11} + \alpha_{12}\bar{f}$.

## G.3 Mixture Model Inversion

Given beliefs $\{r^e_{zt}\}$, the H-type and N-type choice probabilities are

$$A^H_{zt} = \Lambda\big(\alpha_1(\bar{f}_{zt}) r^e_{zt} + \Delta(x^H_{zt}) + s_{zt}\big),$$
$$A^N_{zt} = \Lambda\big(\Delta(x^N_{zt}) + s_{zt}\big),$$

| | $r_{t-1} = 0$ | | $r_{t-1} = 1$ | |
|---|---|---|---|---|
| Weekday | Coef. | SE | Coef. | SE |
| *Panel A: Baseline Intercepts* | | | | |
| Monday | -1.01 | (2.44) | -3.86 | (1.60) |
| Tuesday | -2.41 | (2.49) | 0.13 | (5.56) |
| Wednesday | 0.01 | (2.92) | -1.21 | (2.99) |
| Thursday | 0.59 | (2.63) | -3.19 | (3.89) |
| Friday | (ref.) | | -9.10 | (4.13) |
| Saturday | -14.82 | (6.19) | -5.44 | (1.62) |
| Sunday | 3.10 | (1.68) | -5.40 | (2.27) |
| *Panel B: Hispanic Foreign-Born Share Interactions* | | | | |
| Monday | -1.21 | (0.79) | 0.00 | (0.00) |
| Tuesday | 0.29 | (0.71) | 0.59 | (2.26) |
| Wednesday | 0.05 | (0.77) | 1.98 | (1.43) |
| Thursday | -0.48 | (0.74) | 2.46 | (1.86) |
| Friday | 1.86 | (0.69) | 4.02 | (0.80) |
| Saturday | 3.79 | (1.77) | -0.43 | (0.62) |
| Sunday | -2.62 | (0.52) | 0.27 | (0.38) |
| *Panel C: Log Median Household Income Interactions* | | | | |
| Monday | 0.08 | (0.19) | 0.00 | (0.00) |
| Tuesday | 0.23 | (0.16) | 0.04 | (0.46) |
| Wednesday | -0.00 | (0.24) | 0.16 | (0.27) |
| Thursday | -0.06 | (0.20) | 0.33 | (0.34) |
| Friday | -0.03 | (0.15) | 0.80 | (0.43) |
| Saturday | 1.21 | (0.54) | 0.13 | (0.10) |
| Sunday | -0.41 | (0.10) | 0.13 | (0.17) |
| AOR Fixed Effects | Yes (16 districts) | | | |
| Observations | 36,608 | | | |
| Log-likelihood | -6158.4 | | | |

**Table A2:** Probit Regression for Prior Belief Initialization

*Notes:* Probit regression of raid indicator on weekday $\times$ previous-day-raid fixed effects interacted with ZCTA demographics. Sample restricted to first 105 days (burn-in period) for ZCTAs in the top quintile of Hispanic foreign-born share with at least 30 raids. $r_{t-1} = 0$ indicates no raid on the previous day; $r_{t-1} = 1$ indicates a raid occurred. Standard errors in parentheses, two-way clustered by ZCTA and date.

where $\Lambda(\cdot)$ is the logistic CDF and $s_{zt}$ is the structural shock. The observed economic activity rate is the mixture

$$A_{zt} = h_z \cdot A_{zt}^H + (1 - h_z) \cdot A_{zt}^N.$$

**Inversion procedure.** Because $s_{zt}$ enters both $\Lambda(\cdot)$ terms additively and $\Lambda$ is strictly increasing, $A_{zt}$ is strictly increasing in $s_{zt}$ with range $(0, 1)$. Thus, any observed $A_{zt}$ admits a unique $s_{zt}$ that rationalizes it. For each observation $(z, t)$, we seek the shock $s_{zt}$ that rationalizes the observed $A_{zt}$, which we find by solving the fixed-point problem with a simple bisection algorithm.

We proceed as follows:

1. Initialize pent-up states: $x^H_{z,t_0} = x^N_{z,t_0} = 0$.
2. For each $t = t_0 + 1, \ldots, T$:
   (a) Given current states $(x^H_{zt}, x^N_{zt})$, use bisection to find $s_{zt} \in [-30, 30]$ such that the mixture probability equals $A_{zt}$.
   (b) Update states:
$$x^\tau_{z,t+1} = (1-\delta)x^\tau_{zt} + \kappa - p^\tau_{zt} \tag{24}$$
3. Output the sequence of recovered shocks $\{s_{zt}\}$.

The bounds $s \in [-30, 30]$ ensure numerical stability; in practice, recovered shocks rarely approach these bounds.

## G.4 Fixed Effects Concentration

The structural shocks $s_{zt}$ should be orthogonal to the fixed effects structure if the model is correctly specified. We exploit this by concentrating out the fixed effects through regression. This is computationally convenient because our model has 1339-105 = 1234 time fixed effects, and $352 \times 7 = 2464$ ZCTA-by-weekday fixed effects.

**Regression specification.** We regress the recovered shocks on date and ZCTA-by-weekday fixed effects:

$$s_{zt} = \xi_t + \zeta_{z,d(t)} + \varepsilon_{zt}. \tag{25}$$

The date fixed effects $\xi_t$ absorb aggregate shocks (holidays, macroeconomic conditions). The ZCTA-by-weekday fixed effects $\zeta_{z,d(t)}$ absorb persistent within-week consumption patterns.

**Objective function.** The estimation objective is the residual sum of squares:

$$Q(\boldsymbol{\theta}) = \sum_{z,t} \hat{\varepsilon}^2_{zt} = \sum_{z,t} \big(s_{zt}(\boldsymbol{\theta}) - \hat{\xi}_t - \hat{\zeta}_{z,d(t)}\big)^2. \tag{26}$$

## G.5 Optimization

**Algorithm.** We minimize $Q(\boldsymbol{\theta})$ using L-BFGS-B (Byrd et al., 1995), a quasi-Newton method that accommodates box constraints. The parameter bounds are:

- $\psi \in [0.0001, 10]$ (pent-up cost)
- $\delta \in [0.02, 0.98]$ (depreciation rate)
- $\alpha_{11} \in [-20, 20]$ (belief effect)
- $\alpha_{12} \in [-20, 20]$ (interaction coefficient)

To guard against local optima, we employ a multi-start optimization with 24 random initial points drawn uniformly from the parameter bounds.

Figure A15 displays the profiled objective function for each parameter, holding the others fixed at their estimated values. The profiles are well-behaved with clear global minima, confirming that our estimates correspond to a unique optimum.

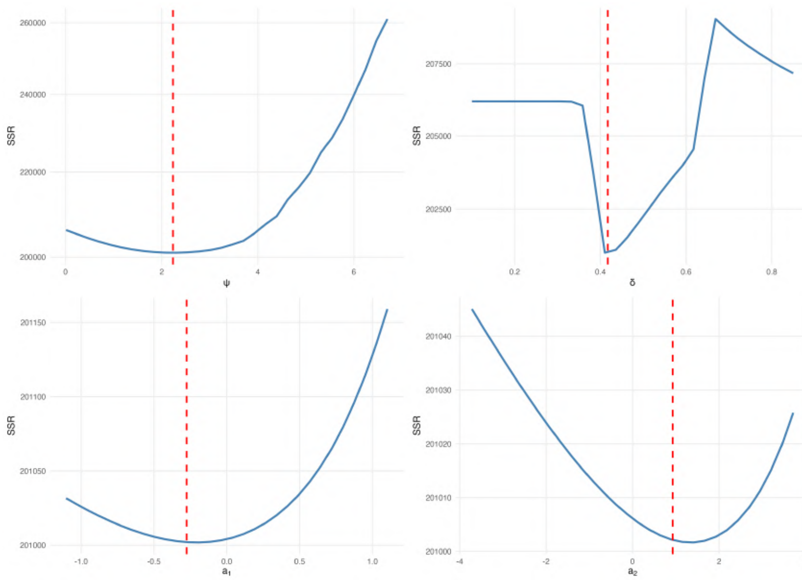



**Figure A15:** Profile Likelihood for Structural Parameters

*Notes:* Each panel shows the objective function $Q(\boldsymbol{\theta})$ as a function of one parameter, holding the remaining parameters fixed at their estimated values. Vertical dashed lines indicate the parameter estimates. The well-behaved profiles with clear global minima indicate that our estimates correspond to a unique optimum.

## G.6 Standard Errors

We compute standard errors in two steps.

**Numerical Hessian.** We approximate the Hessian matrix $H = \nabla^2 Q(\hat{\boldsymbol{\theta}})$ using central finite differences with step size $10^{-4}$.

**HAC adjustment.** Because the recovered shocks may exhibit serial correlation and heteroskedasticity, we compute heteroskedasticity and autocorrelation-consistent (HAC) standard errors using the Newey-West estimator (Newey and West, 1987). We report results across a range of lag specifications to assess robustness.

The asymptotic variance is

$$\widehat{\mathrm{Var}}(\hat{\boldsymbol{\theta}}) = H^{-1}\hat{\Omega}H^{-1}, \tag{27}$$

where $\hat{\Omega}$ is the HAC-adjusted outer product of the gradient.

Figure A16 displays the autocorrelation function of the recovered residuals, showing persistence at short lags that motivates the HAC adjustment.

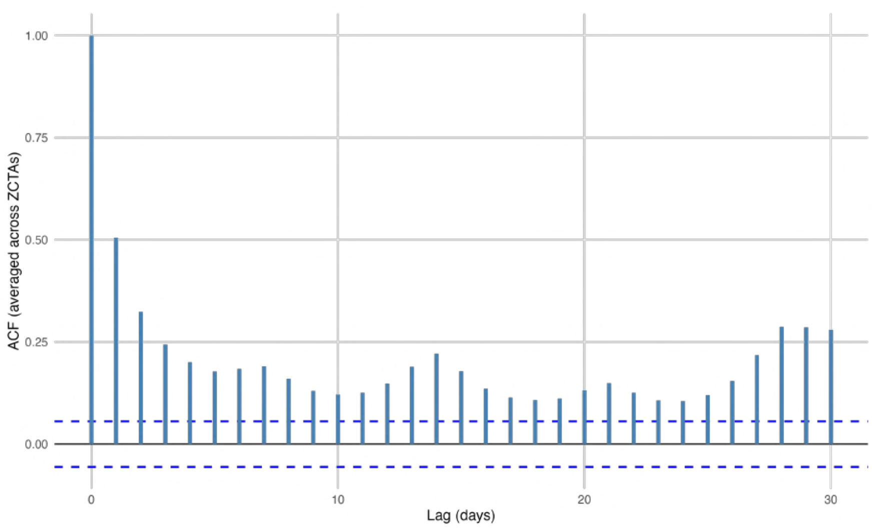
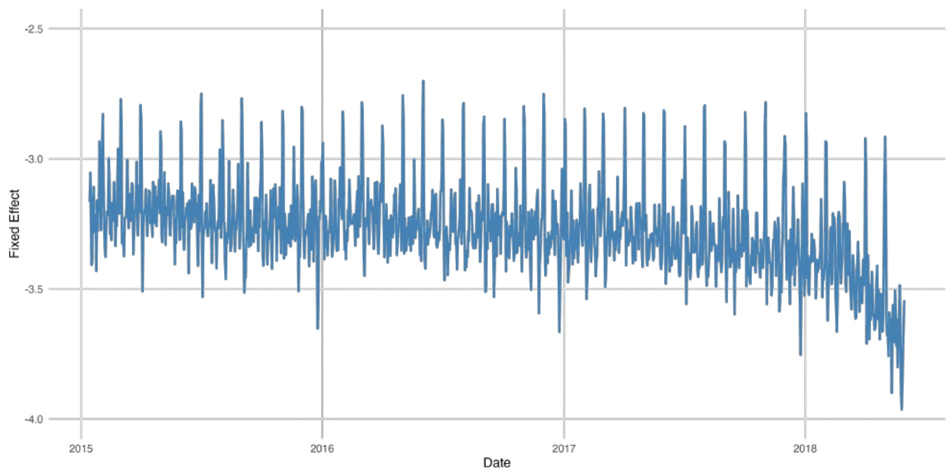



**Figure A16:** Autocorrelation Function of Residuals

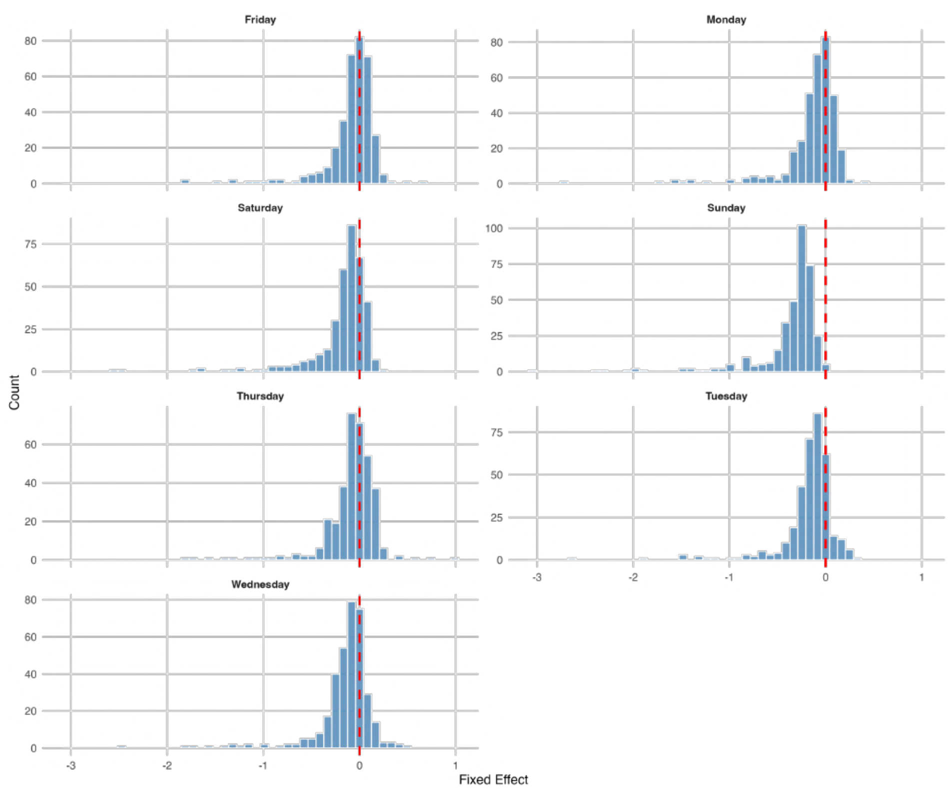
*Notes:* Autocorrelation function of the recovered structural shocks $\hat{s}_{zt}$ from the mixture model inversion. The persistence at short lags motivates the use of Newey-West HAC standard errors. Dashed lines indicate 95 percent confidence bands under the null of no autocorrelation.

## G.7 Estimated Fixed Effects

Figure A17 displays the estimated fixed effects from the structural estimation.



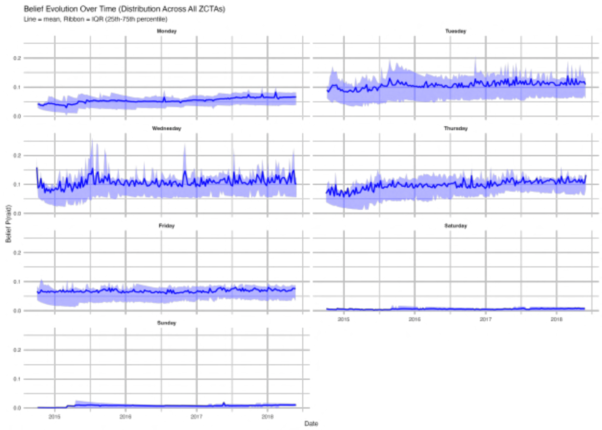

**(a)** Date fixed effects over time




**(b)** Distribution of ZCTA-by-weekday fixed effects

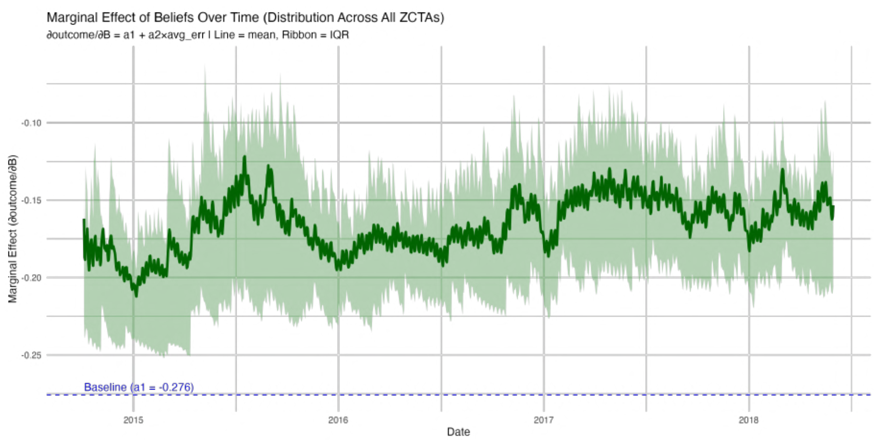
**Figure A17:** Estimated Fixed Effects from Structural Model

*Notes:* Panel (a) shows the estimated date fixed effects $\hat{\eta}_t$ over the sample period, capturing aggregate shocks to economic activity (holidays, macroeconomic conditions). Panel (b) shows the distribution of ZCTA-by-weekday fixed effects $\hat{\mu}_{z,d}$, which absorb persistent within-week consumption patterns at the community level. The estimation includes 1,234 date fixed effects and 2,464 ZCTA-by-weekday fixed effects.




**(a)** Distribution of beliefs by weekday


**(b)** Cumulative forecast error by weekday




**(c)** Marginal effect of beliefs over time

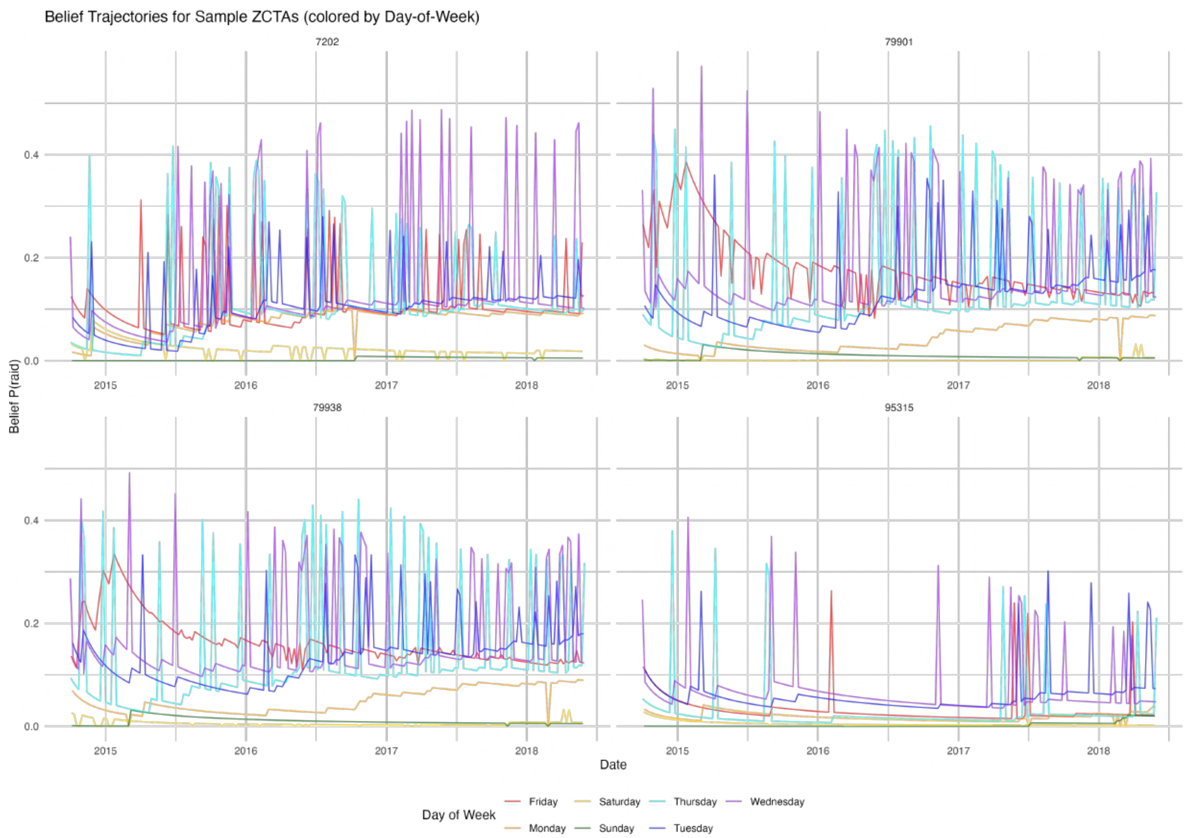
**Figure A18:** Implied Beliefs, Forecast Errors, and Time-Varying Belief Sensitivity

*Notes:* Panel (a) shows the distribution of implied beliefs $B_{zd(t)}$ by weekday across all ZCTAs in the sample, over time. Panel (b) shows the distribution of implied cumulative forecast errors $\bar{f}_{zt}$ by weekday across all ZCTAs in the sample, over time. Panel (c) displays the distribution of the marginal effect of beliefs on economic activity, $\alpha_{11} + \alpha_{12} \cdot \bar{f}_{zt}$, over time, across all ZCTAs in the sample.

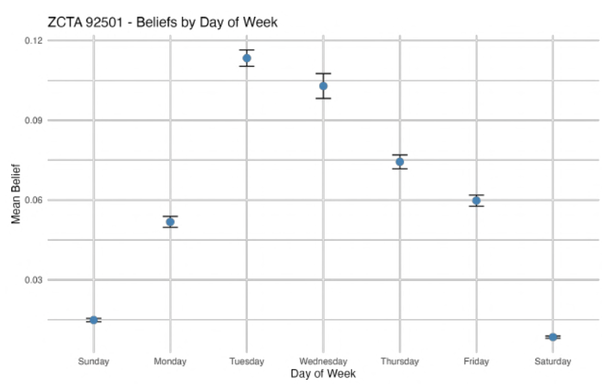
## G.8 Implied Beliefs: Sample Averages and ZCTA Examples

Figure A19 illustrates the implied beliefs from the structural model. Panel (a) shows sample-wide average beliefs by day of week over time. Panels (b) and (c) show belief evolution for two example ZCTAs, illustrating heterogeneity in belief dynamics across communities.

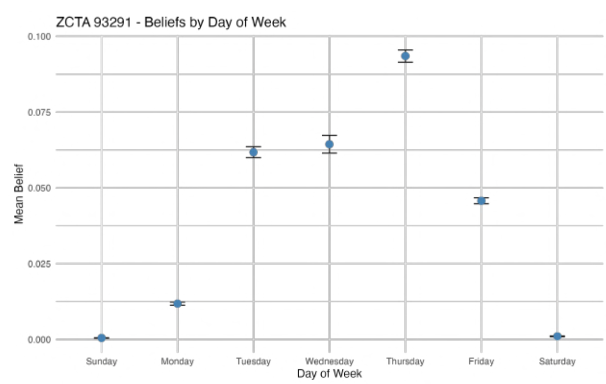
## G.9 Impulse Responses




**(a)** Beliefs by weekday over time for a subset of ZCTAs




**(b)** ZCTA 92501 (Riverside, CA)




**(c)** ZCTA 93291 (Visalia, CA)

**Figure A19:** Implied weekday average beliefs for a subset of ZCTAs

*Notes:* Panel (a) shows the implied belief evolution $B_{zd(t)}$ by weekday over the estimation period for four different ZCTAs. Panels (b) and (c) show the implied mean beliefs by weekday for two example ZCTAs in California.

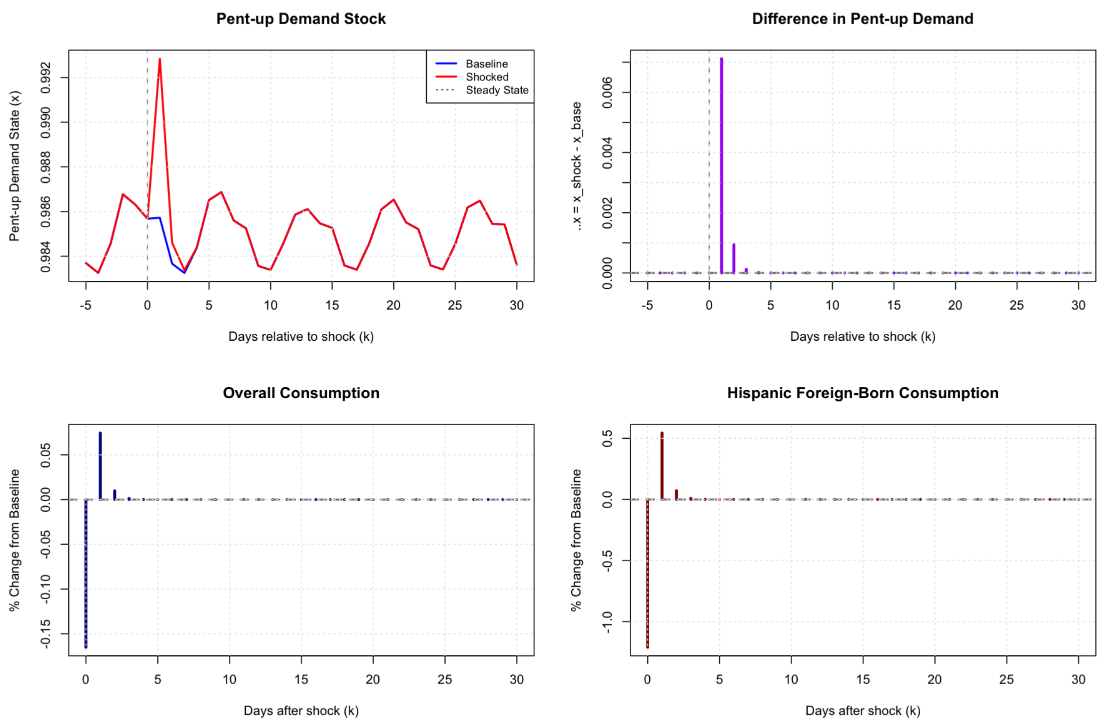



**Figure A20:** Impulse Response to a Belief Shock: Beliefs Evolve

*Notes:* Impulse response to a 15 percentage point belief shock from steady state, for the scenario in which beliefs are allowed to update over time after the shock. This is the companion to Figure 6 in the main text, which shows the immediate-reversion scenario. The figure presents four subfigures: the top left, the evolution of the pent-up demand stock; the top right, the day-to-day changes in the pent-up demand stock; the bottom left, the daily percent changes in economic activity overall; and the bottom right, the daily percent changes in economic activity among the Hispanic foreign born.

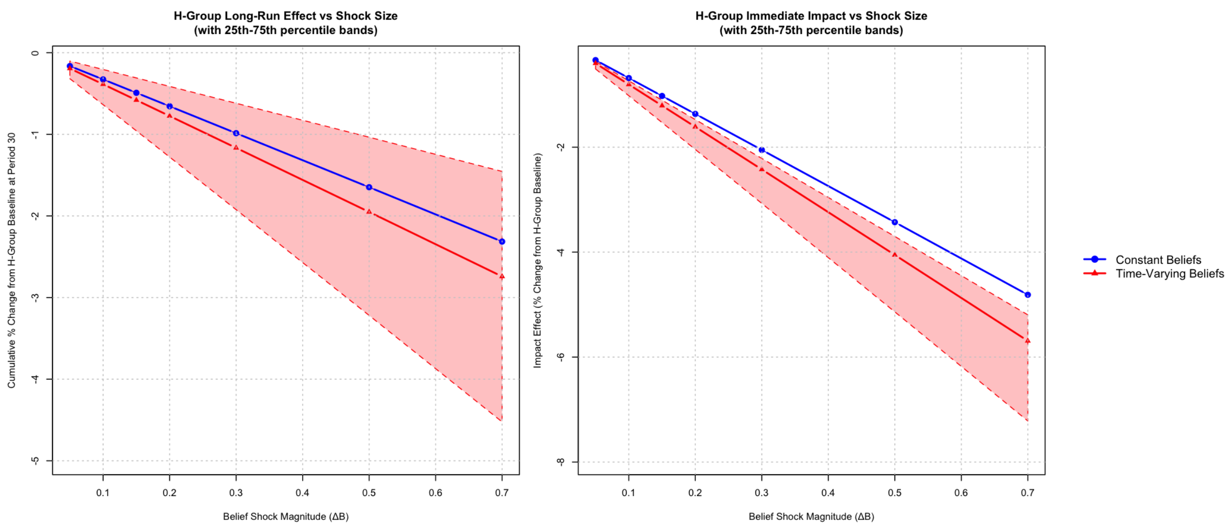
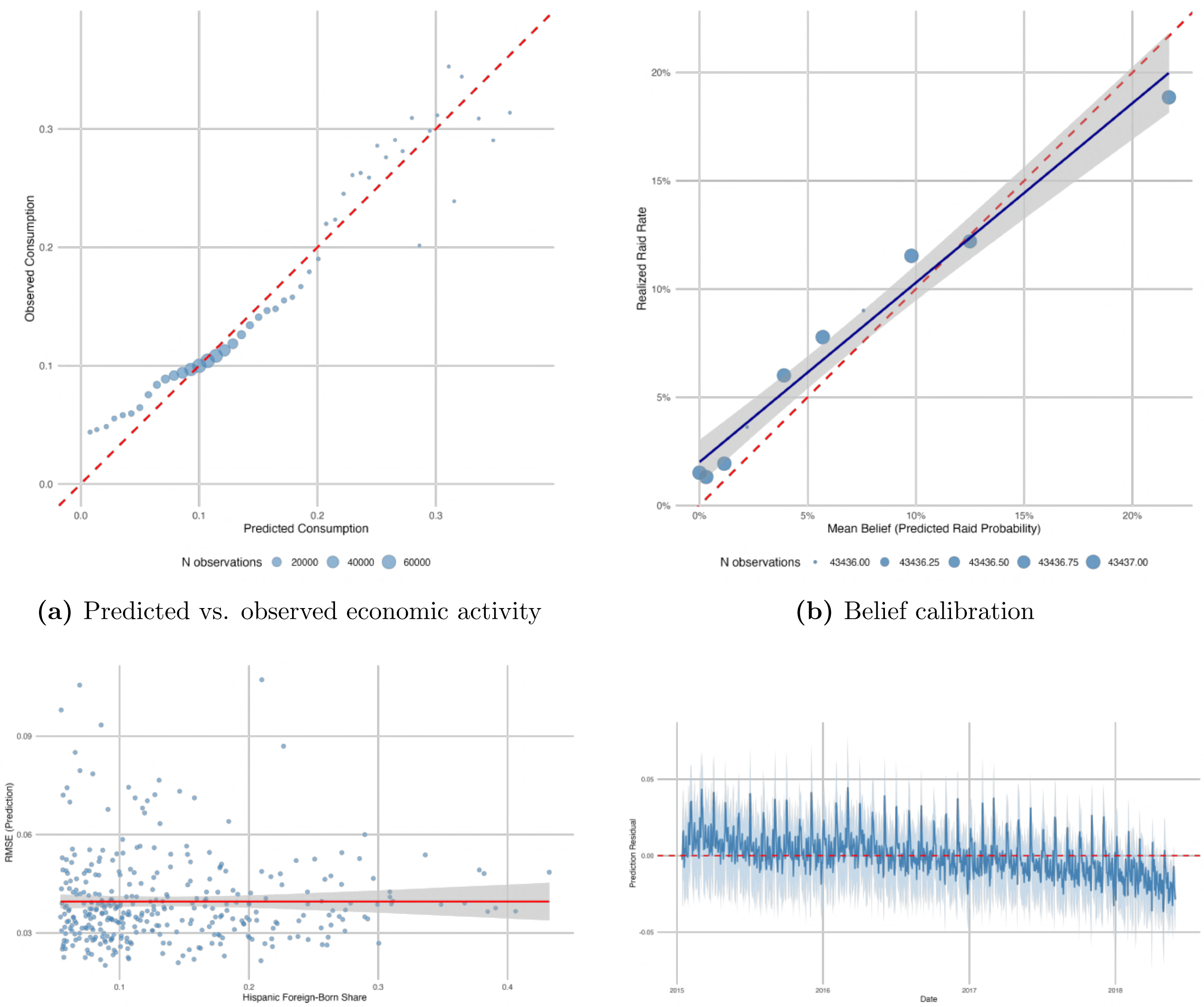



**Figure A21:** Nonlinearity in Impulse Responses: H-Type Group

*Notes:* Impulse response effects on economic activity as a percent of the steady state baseline for the Hispanic foreign born group under belief shocks of magnitudes varying between 5 and 70 percentage points. The left-hand-side panel presents the net cumulative effects. The right-hand-side panel presents the effect on impact. The blue lines measure the mean effects under the scenario where beliefs revert to baseline immediately after the shock. The red lines measure the mean effects under the scenario where beliefs are allowed to evolve endogenously after the shock. The pink-shaded areas represent the IQR of the distribution of impulse-response effects across ZCTAs for the scenario where beliefs can evolve endogenously.

# H Model Validation




**(a)** Predicted vs. observed economic activity

**(b)** Belief calibration

**(c)** Model fit by Hispanic foreign-born share

**(d)** Residuals over time

**Figure A22:** Model Fit Diagnostics

*Notes:* Panel (a) shows a binned scatter plot of predicted versus observed economic activity (share of accounts active), with 100 equal-sized bins. Panel (b) shows belief calibration: binned mean beliefs plotted against observed raid frequencies. Panel (c) shows RMSE by decile of Hispanic foreign-born share. Panel (d) shows residuals over time, illustrating time-series fit.

Figure A23 presents the results of a placebo test for the structural breaks validation. We generate 233 “fake” structural breaks by randomly reassigning break dates within each ZCTA’s sample period, preserving the overall frequency and ZCTA distribution of breaks. The figure overlays the event study coefficients for real breaks (green line) against placebo breaks (grey line). While forecast errors spike sharply after real breaks, the placebo event study is flat—placebo breaks show no effect on forecast errors. This confirms that the forecast error spikes we observe are not artifacts of our estimation procedure but reflect genuine belief-outcome mismatches caused by actual regime changes.

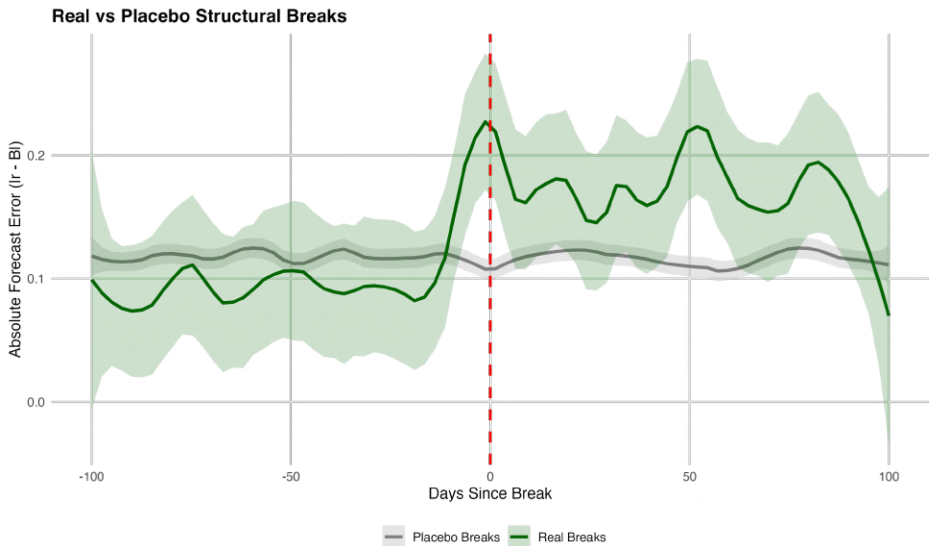



**Figure A23:** Placebo Test: Real vs. Fake Structural Breaks

*Notes:* Event study of mean absolute forecast errors around detected structural breaks. The green line shows the conditional mean of the forecast errors from event time based on detected structural breaks following the Bai and Perron (2003) methodology. The grey line shows the conditional mean of the forecast errors from event time based on placebo structural breaks generated by randomly reassigning break dates within each ZCTA. The shaded regions show 95 percent confidence intervals.

| | (1) Clustering | (2) Support Ratio | (3) log(SCI) | (4) Cohesion PC1 |
|---|---|---|---|---|
| Cohesion measure | -6.15 | -0.70 | -0.042 | -0.042 |
| | (2.51) | (0.27) | (0.034) | (0.017) |
| Observations | 349 | 349 | 349 | 349 |
| $R^2$ | 0.41 | 0.41 | 0.40 | 0.41 |

**Table A3:** Structural Model Fit and Social Cohesion

*Notes:* The dependent variable in all columns is the per-ZCTA RMSE from the structural model's share accounts active residuals. Each column reports the coefficient on a different social cohesion measure: Column 1 uses the clustering coefficient from Chetty et al. (2022), measuring the average across Facebook subscribers in the ZCTA of the share of an individual's Facebook friend pairs who are also friends with each other; column 2 uses the support ratio from Chetty et al. (2022), measuring share of Facebook friendships in the ZCTA that are supported (have at least one mutual friend); column 3 uses the Social Connectedness Index from Bailey et al. (2018), that measures the ratio of Facebook friendships observed between individuals in a given ZCTA as a scaled fraction of all potential friendships between them; column 4 is the first principal component of the three measures. All specifications include log population and log number of raids in the ZCTA as controls. Standard errors in parentheses.

# I Counterfactuals

Figure A24 shows the distribution of consumption losses across ZCTAs for each raid increase scenario. The distributions shift left as the enforcement rate increases, with losses becoming more concentrated at higher negative levels. The distribution of losses is bimodal, and spreads out as the increase in enforcement intensity becomes higher.

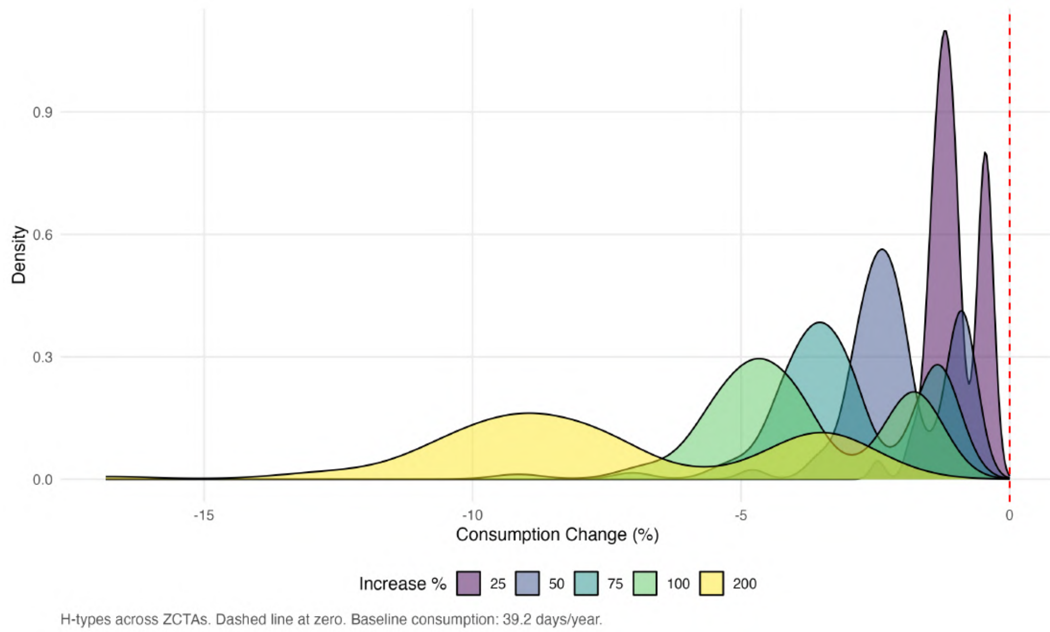
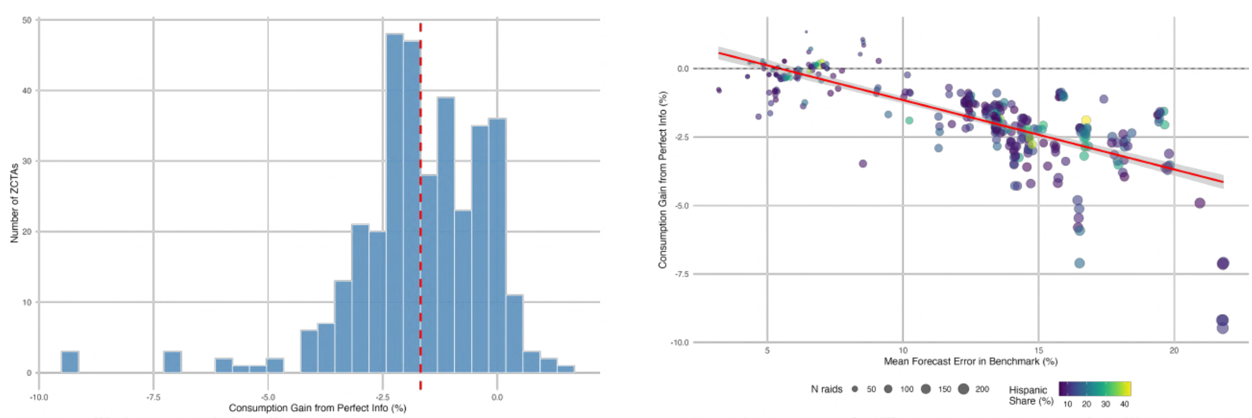



**Figure A24:** Distribution of Consumption Losses by Raid Increase Level

*Notes:* Distribution of consumption losses (percent) among the Hispanic foreign born across the 352 ZCTAs under each raid increase scenario. Losses are computed relative to the baseline (0 percent reduction) and averaged over 50 Monte Carlo simulation draws.

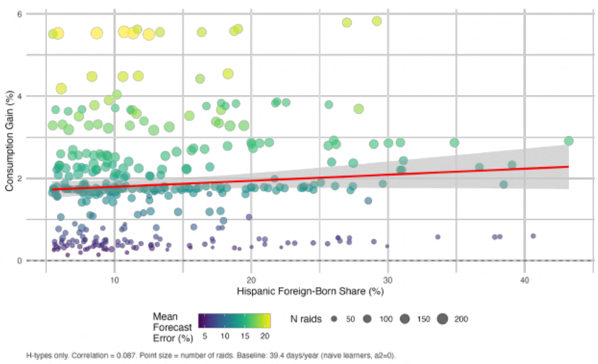



**(a)** Distribution of consumption changes

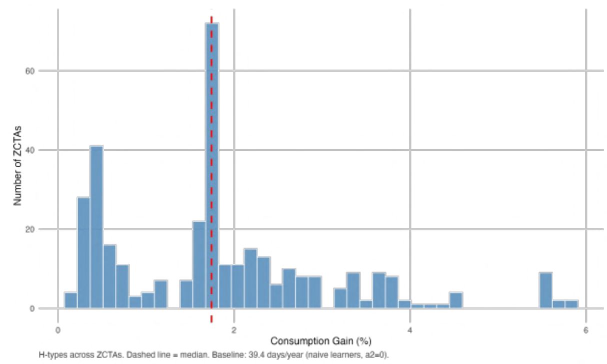
**(b)** Consumption change vs. forecast error

**Figure A25:** Perfect Information Counterfactual

*Notes:* Panel (a) shows the distribution of consumption changes as a fraction of baseline consumption among the Hispanic foreign born under perfect information ($B_{zt} = r_{zt}$, $\bar{f}_{zt} = 0$) relative to the benchmark with Bayesian learning. The mean effect is $-1.7$ percent; 89 percent of ZCTAs experience consumption decreases. Panel (b) plots the corresponding consumption changes across ZCTAs against the model implied average cumulative forecast error. Dot colors represent the Hispanic foreign born share.



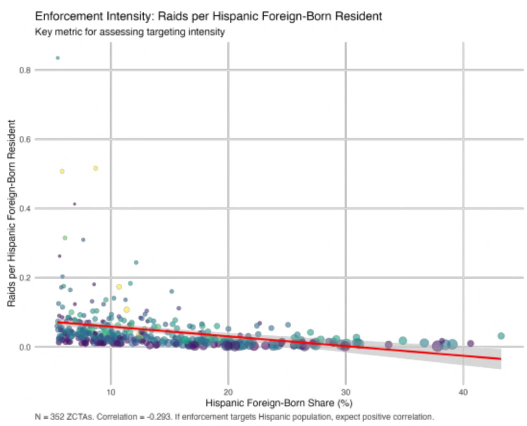

**(a)** Gains vs. Hispanic foreign-born share



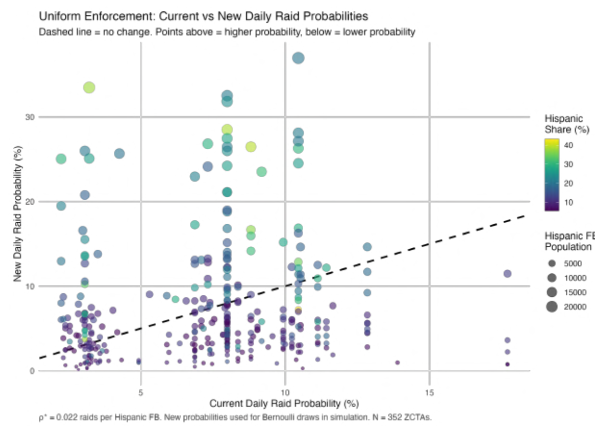

**(b)** Distribution of consumption gains

**Figure A26:** The Value of Sophistication in Learning

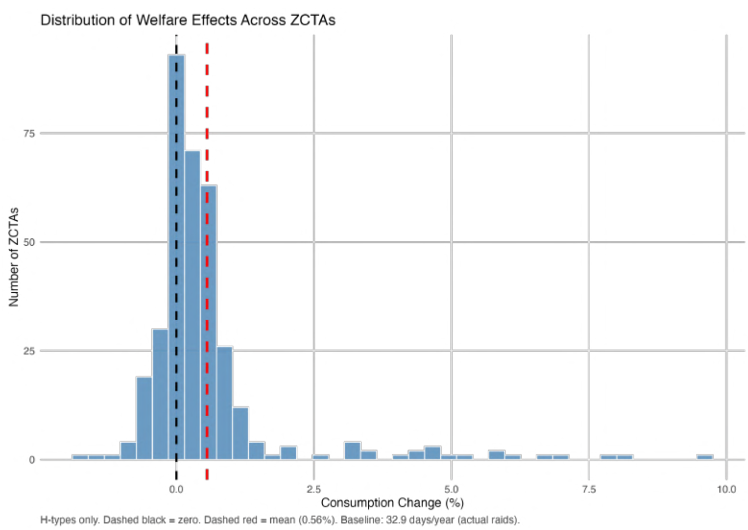
*Notes:* Consumption gains from sophisticated learning ($\hat{\alpha}_{12} = 0.93$) relative to naive Bayesian updating ($\alpha_{12} = 0$). Panel (a) plots percent gains among the Hispanic foreign born against the Hispanic foreign-born share. Dot colors denote average forecast error. Panel (b) shows the distribution of percent gains among the Hispanic foreign born across ZCTAs. The mean gain is 1.85 percent (0.72 days/year).




**(a)** Current enforcement intensity




**(b)** Redistribution of raid probabilities




**(c)** Distribution of welfare gains

**Figure A27:** Uniform Enforcement Counterfactual

*Notes:* Panel (a) shows current enforcement intensity (raids per Hispanic foreign-born resident) against the Hispanic foreign born share. Dot colors denote the number of raids. Panel (b) shows the redistribution of enforcement intensity: observed daily raid probabilities (x axis) against counterfactual probabilities under uniform enforcement intensity (y-axis). Points above the 45-degree line gain raids; points below lose raids. Dot colors denote the Hispanic foreign born share. Panel (c) shows the distribution of consumption gains from uniform enforcement intensity. The mean gain is 0.6 percent (0.2 days/year), with 75 percent of ZCTAs experiencing positive gains.